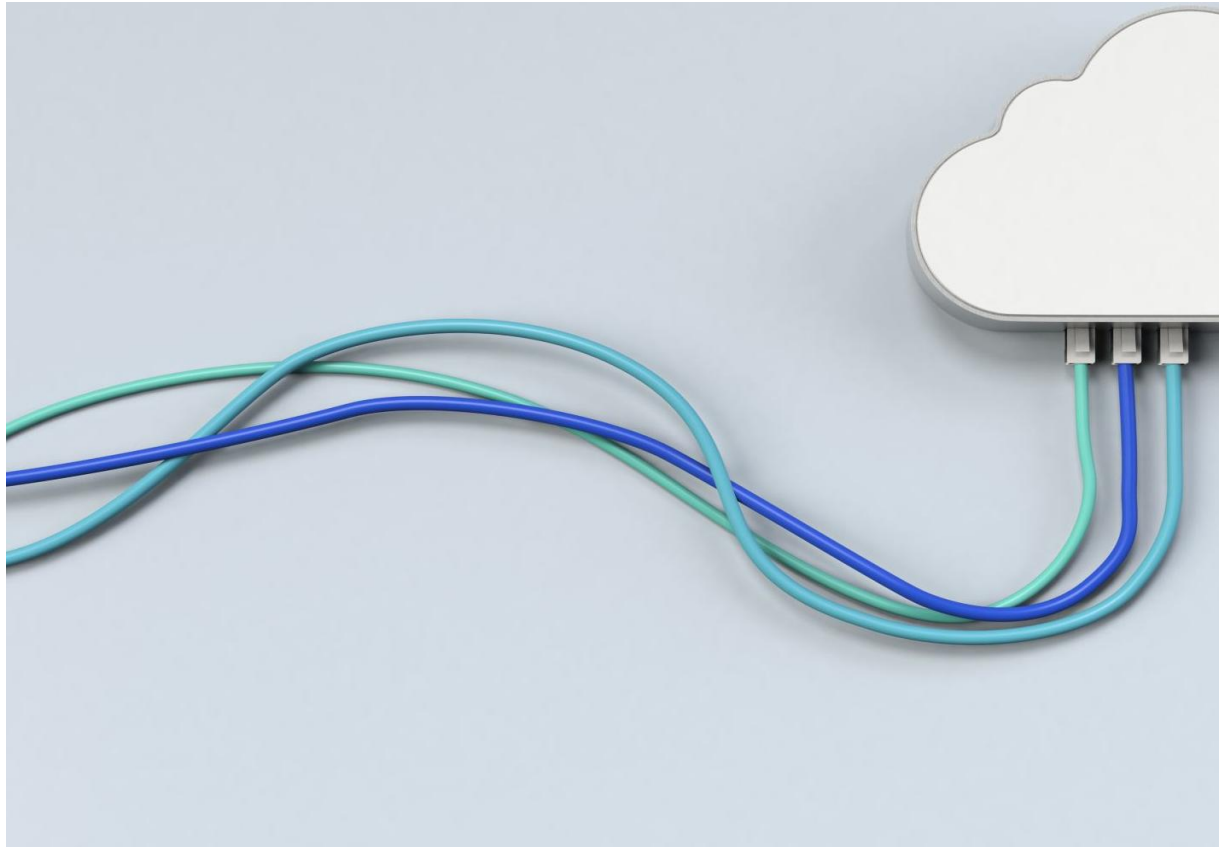


CLOUD COMPUTING

CC-26 BTECH(CSE-DS)





CONTENTS

- **What is the cloud computing?**
- **Roots of cloud computing**
- **Cloud Computing Components**
- **Characteristics**
- **Service Models**
- **Deployment Models**
- **Architecture**
- **Cloud Engineering**
- **Security and Privacy**
- **Cloud Clients**
- **Advantages**

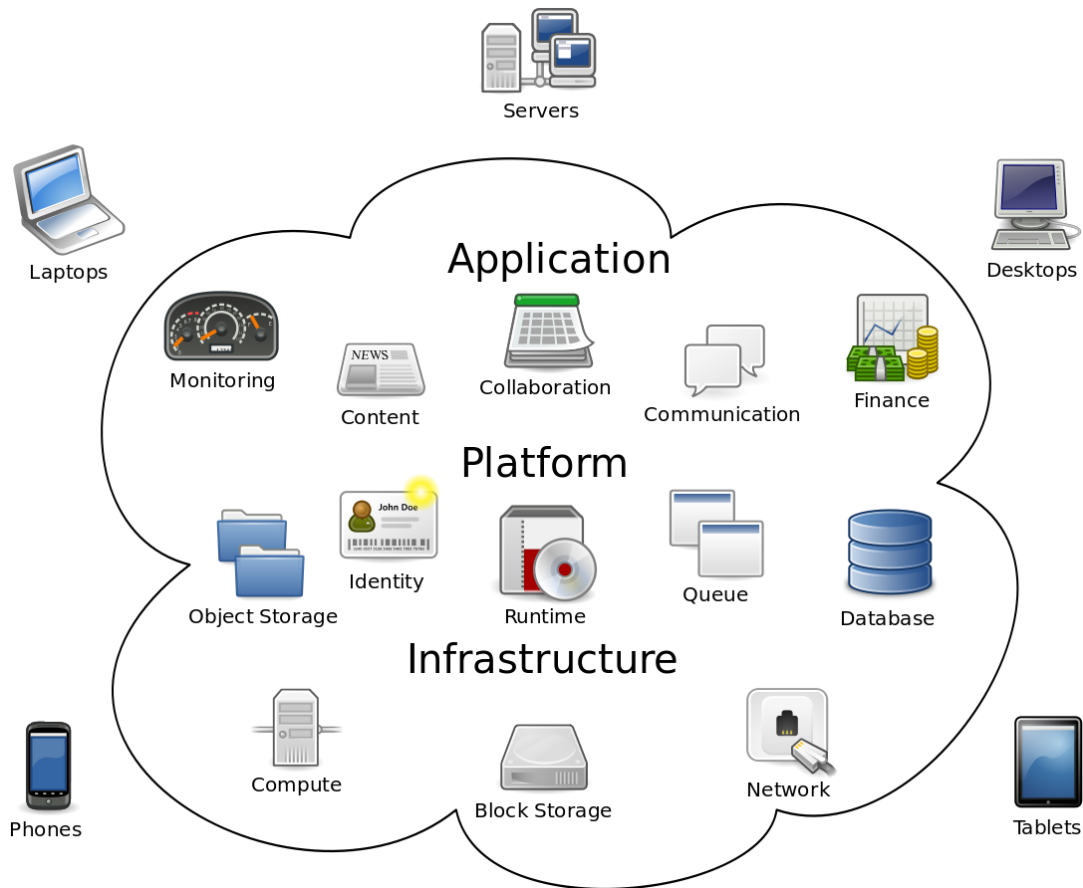
WHAT IS CLOUD COMPUTING?

-
- Cloud computing :
 - Internet-based computing
 - large groups of remote servers are networked
 - sharing of data-processing tasks, centralized data storage, computer services and resources
 - Any computer related task that is done entirely on Internet

WHAT IS CLOUD COMPUTING?

Allows users to deal with the software without having the hardware.

Everything is done by remotely, nothing is saved locally.



CLOUD COMPUTING COMPONENTS

CHARACTERISTICS

Empowerment

Agility and API

Cost and
Security

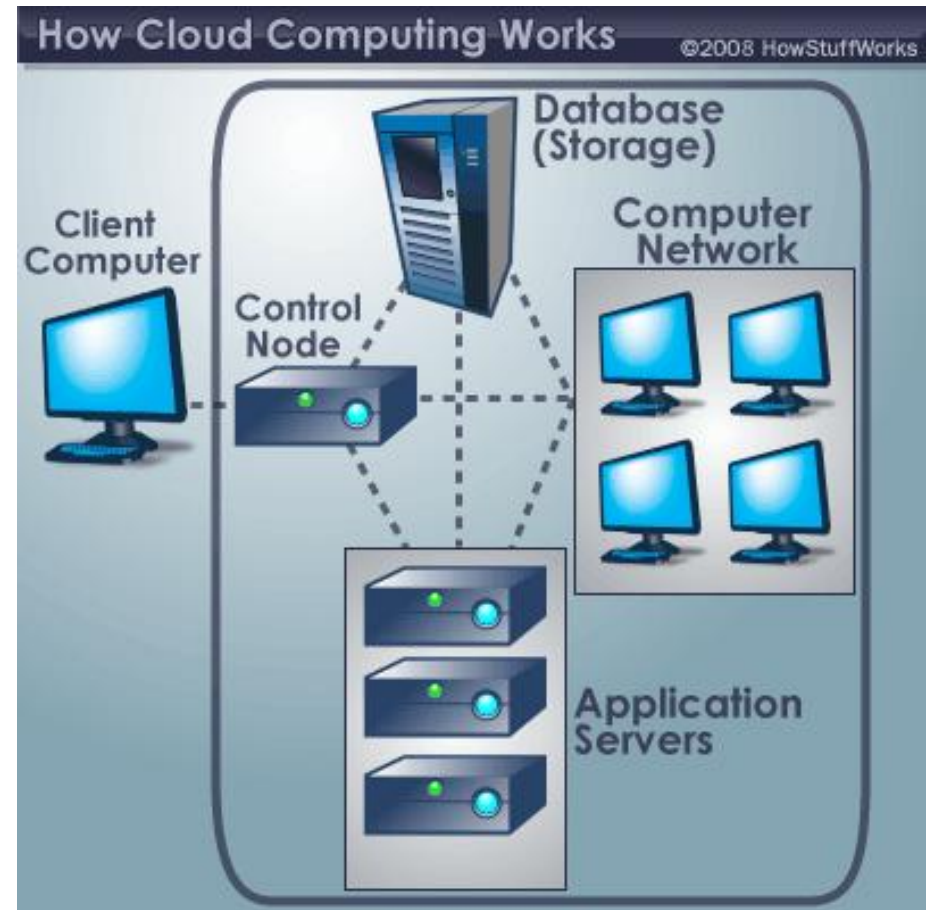
Device and
location
independence

Visualization

Multi-tenancy

Reliability and
Maintenance

Scalability and
Elasticity



How Cloud Computing Works?

Primary Terminologies

- **Mainframe Era:** The Mainframe Era was famous in the 1950s–1970s times. and it was a large central computer that was used by many users through terminals.
- **Service-Oriented Architecture (SOA):** It's a type of design approach where applications are built as smaller, independent pieces called services. and after that, they will be integrated and shared over the network.
- **Grid Computing:** Grid computing is basically combining many computers together to solve big and complex tasks faster.
- **Application Service Providers (ASPs):** It's like early versions of cloud services that provide applications or services that you could use through the internet.
- **Cloud Computing:** It is an on-demand self-service model that basically provides a pool of resources such as networks, computing **pay-as-you-go**, storage, databases, and many more services within a few seconds with minimal management and without human interaction. Another significant advantage is the pay-as-you-go model and time to market.

Traditional Computing	Cloud Computing
Data storage on local Internal / hard drive	Data storage in the cloud or external server
Extra storage – pay for hardware, no unlimited storage	Unlimited storage – no extra hardware required
For hardware configuration, you may hire expert	No need to do hardware configuration – dynamic resources
Pay for power, electricity, AC cooling etc.	No expense on power, electricity, AC cooling etc.
If you need to add new server, it will take few days and more charges apply.	Pay for use, dynamic resources allocation.
Fixed charges – installation, hardware used.	If resources are use less, then pay less
Lack of isolation – high risk of security and poor performance.	No need to worry about security / loss of data.
Backup and recovery take risky.	Backup and recovery very easy.
Extra pay for software installation & upgrade.	Automatic software upgrade available
Pay for extra hardware & other resources	On demand resources

CLOUD COMPUTING ARCHITECTURE

Cloud Computing architecture comprises of many cloud components, which are loosely coupled. We can broadly divide the cloud architecture into two parts:

Front End

Back End

- Each of the ends is connected through a network, usually Internet.

Front End

- The **front end** refers to the client part of cloud computing system. It consists of interfaces and applications that are required to access the cloud computing platforms, Example - Web Browser.

Back End

- The **back End** refers to the cloud itself. It consists of all the resources required to provide cloud computing services. It comprises of huge data storage, virtual machines, security mechanism, services, deployment models, servers, etc.

CLOUD INFRASTRUCTURE

It consists of servers, storage devices, network, cloud management software, deployment software, and platform virtualization.

Management Software

Deployment Software

Hypervisor

Network

Server

Storage



Hypervisor

Hypervisor is a **firmware** or **low-level program** that acts as a Virtual Machine Manager. It allows to share the single physical instance of cloud resources between several tenants.



Management Software

It helps to maintain and configure the infrastructure.



Deployment Software

It helps to deploy and integrate the application on the cloud.



Network

It is the key component of cloud infrastructure. It allows to connect cloud services over the Internet. It is also possible to deliver network as a utility over the Internet, which means, the customer can customize the network route and protocol.

Infrastructure

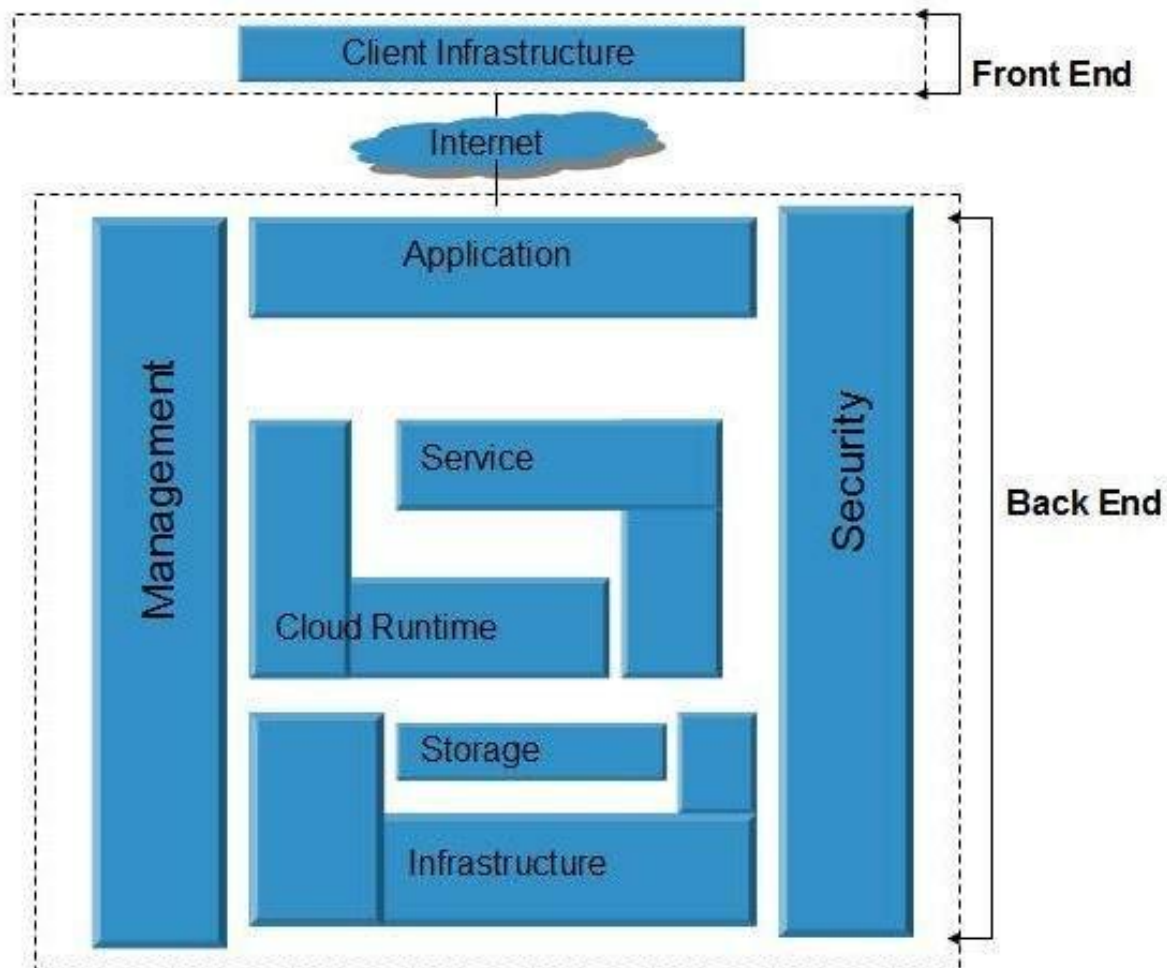


Server

- The **server** helps to compute the resource sharing and offers other services such as resource allocation and de-allocation, monitoring the resources, providing security etc.

Storage

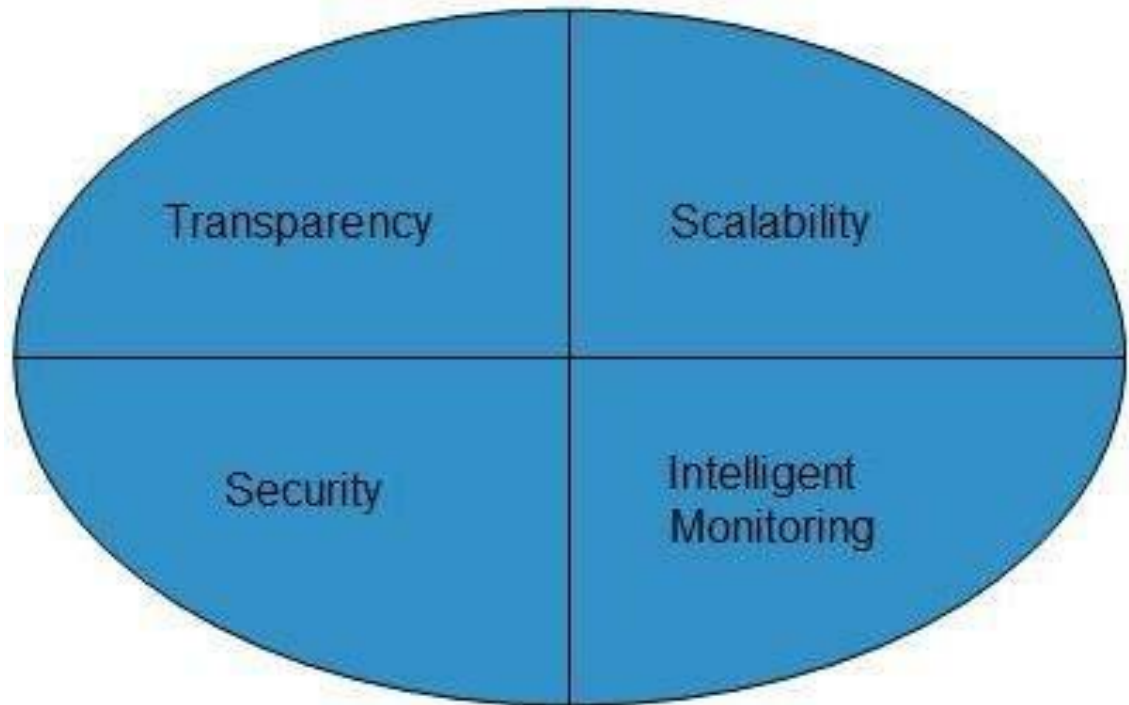
- Cloud keeps multiple replicas of storage. If one of the storage resources fails, then it can be extracted from another one, which makes cloud computing more reliable.



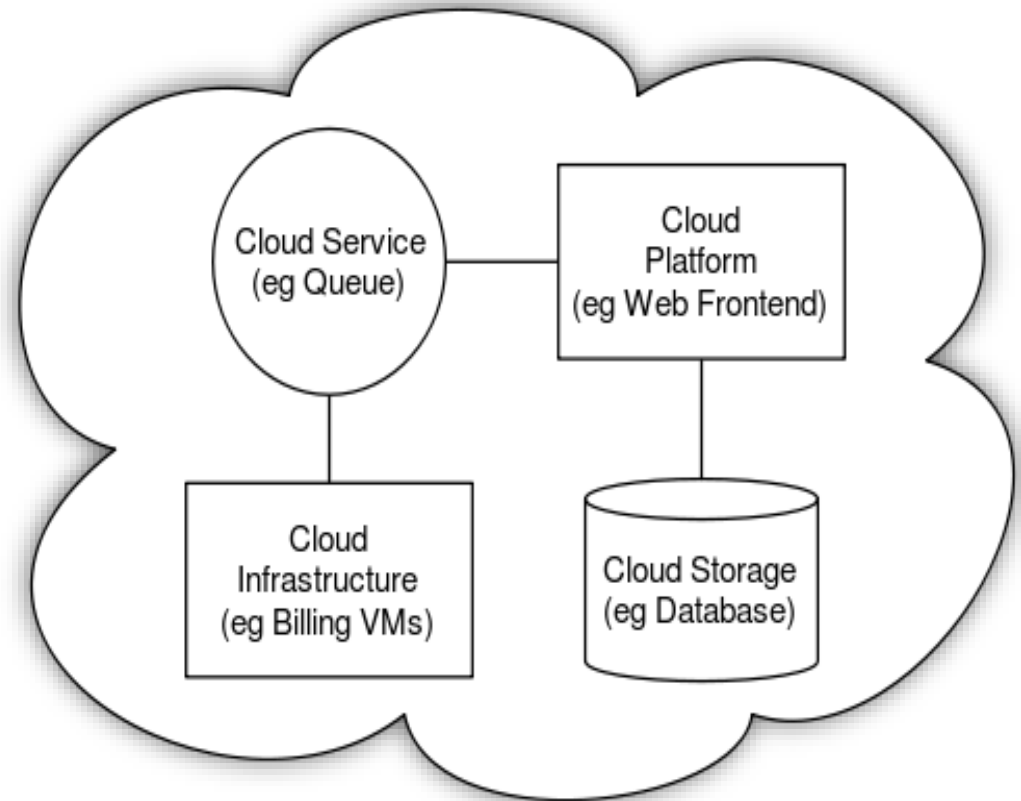
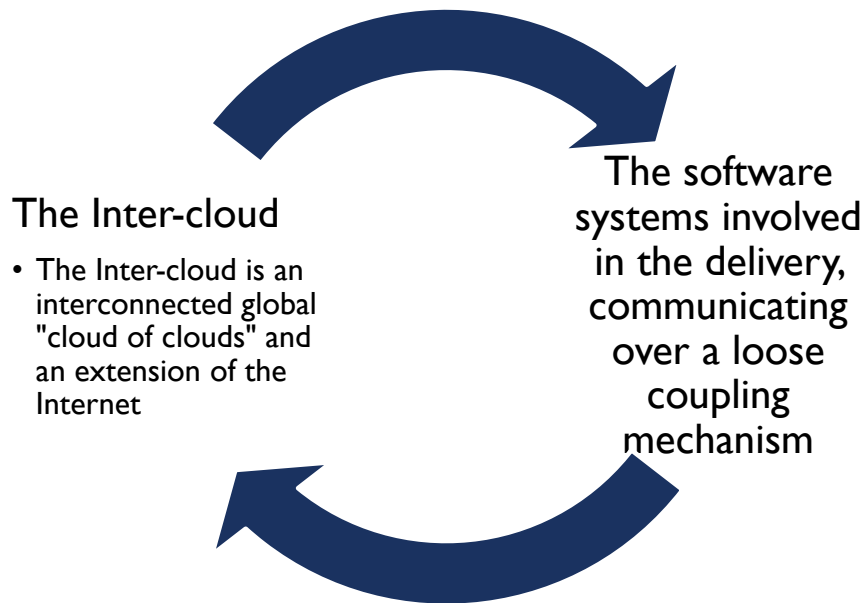
Architecture

- **Infrastructural Constraints:**

Fundamental constraints that cloud infrastructure should implement are shown in the following diagram:



ARCHITECTURE



Cloud Architecture

DEPLOYMENT MODELS



Public cloud

Done by service providers



Community cloud

organizations from a
specific community with
common concerns



Private cloud

operated solely for a single
organization



Hybrid cloud

composition of two or
more clouds (private,
community or public)



Private Cloud Rentals

option to consider when
security is a concern



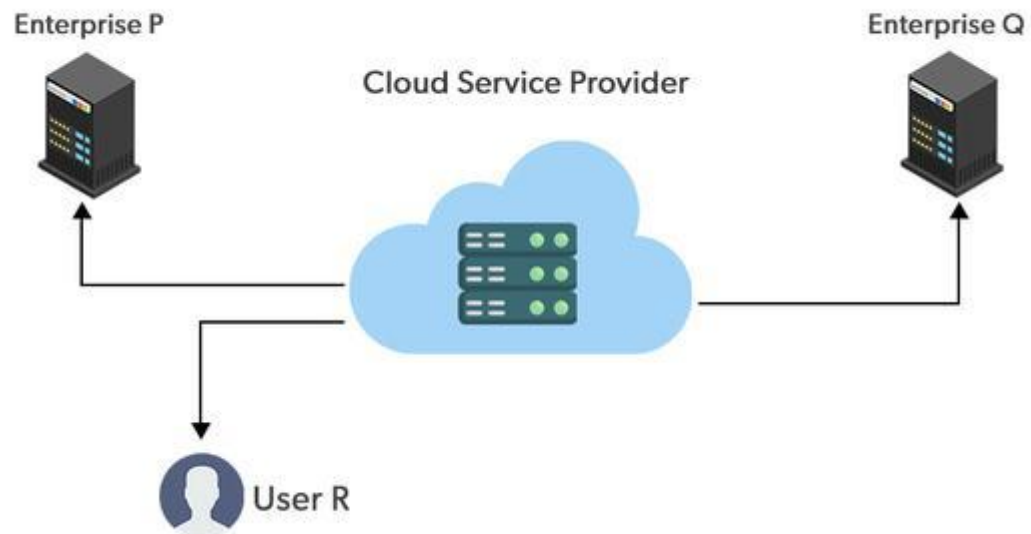
Cloud computing offers deployment models that provides different balances of cost, control, and security for diverse organizational needs.

Deployment Models :

- **Public Cloud:** Resources (servers, storage) owned and operated by third-party providers (like AWS, Azure, GCP) and shared among multiple organizations over the internet, offering high scalability and cost-effectiveness.
- **Private Cloud:** Infrastructure dedicated solely to one organization, either on-premises or managed by a third party, providing greater control and security for sensitive data.
- **Hybrid Cloud:** Combines public and private clouds, allowing data and applications to move between them, leveraging the scalability of public with the security of private.
- **Community Cloud:** Shared by several organizations with common concerns (e.g., security, compliance), providing a collaborative environment with shared infrastructure.

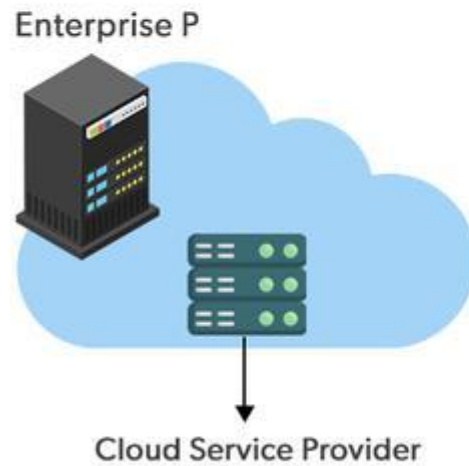
*** Public, Private, Hybrid define where and how the infrastructure is hosted and managed.*

Public Cloud

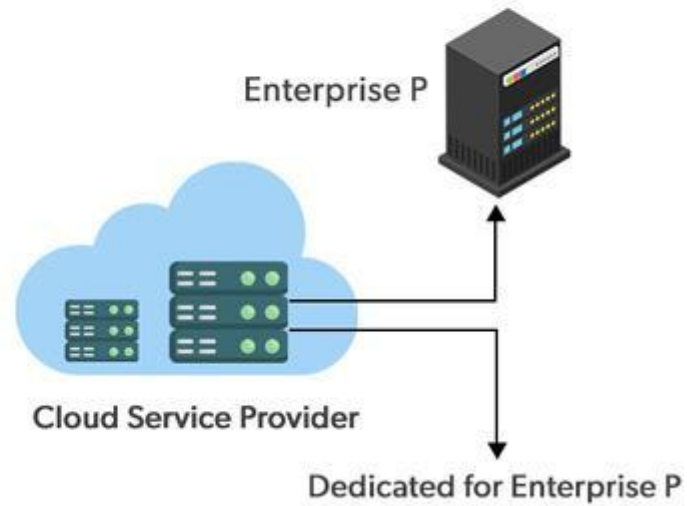


Private Cloud

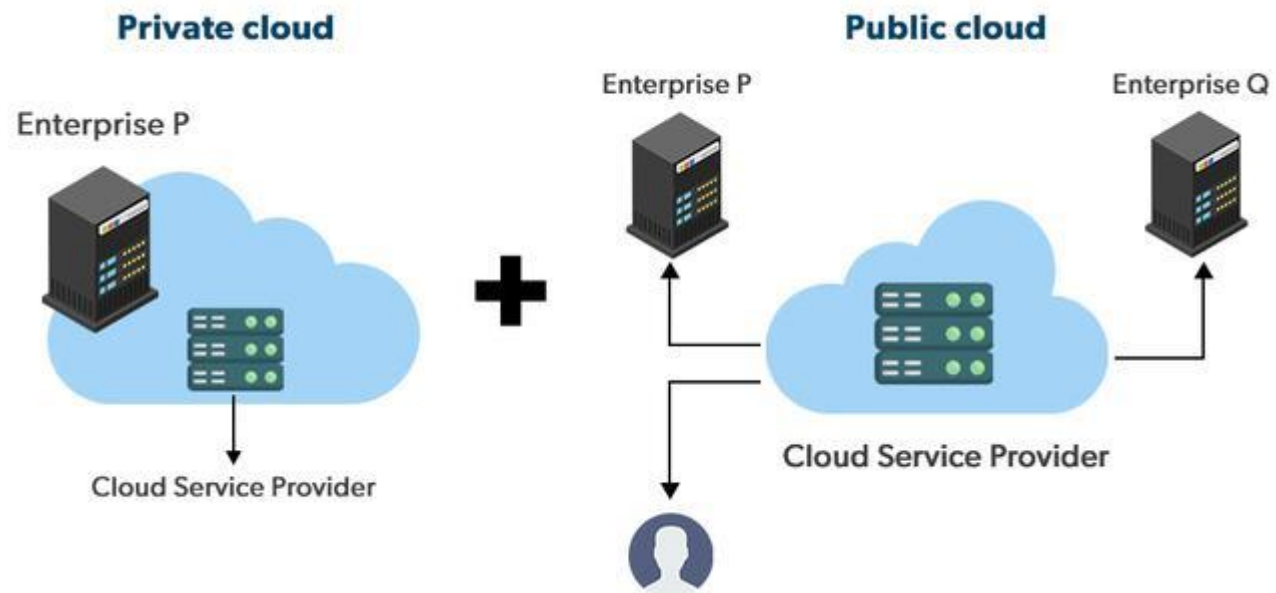
On premise Private cloud



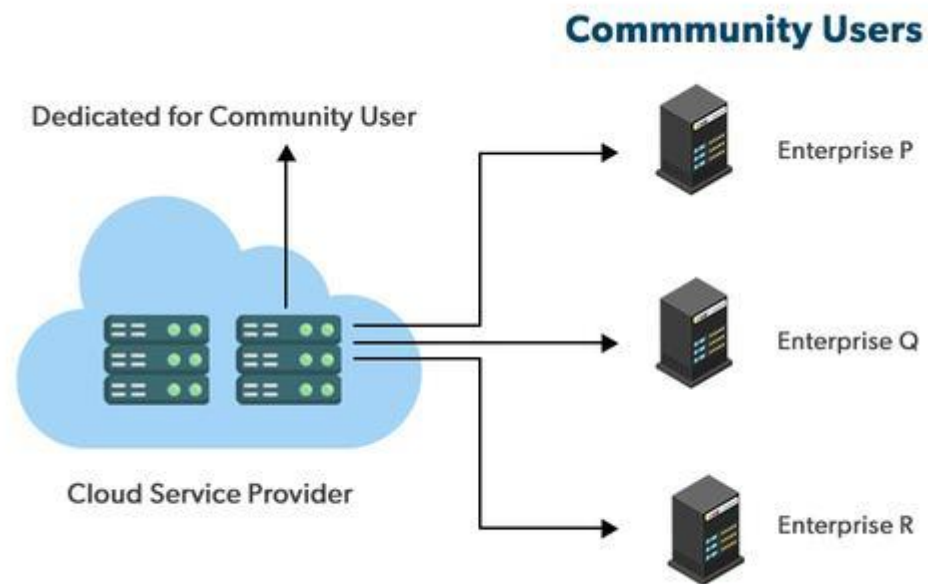
Externally hosted Private cloud



Hybrid Cloud



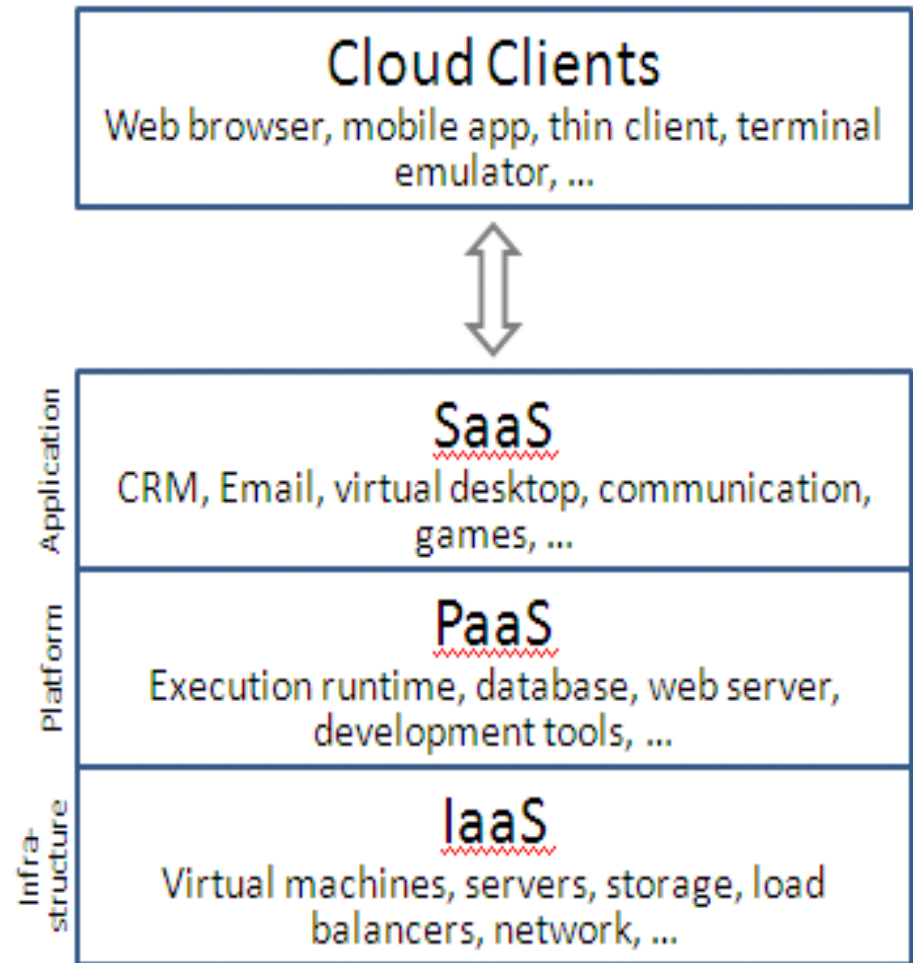
Community Cloud



FACTORS	PUBLIC CLOUD	PRIVATE CLOUD	COMMUNITY CLOUD	HYBRID CLOUD
Initial Setup	Easy	Complex, requires a professional team to setup	Complex, requires a professional team to setup	Complex, requires a professional team to setup
Scalability and Flexibility	High	High	Fixed	High
Cost-Comparison	Cost-Effective	Costly	Distributed cost among members	Between public and private cloud
Reliability	Low	Low	High	High
Data Security	Low	High	High	High
Data Privacy	Low	High	High	High

SERVICE MODELS

- **Infrastructure as a Service (IaaS)**
 - Basic, service users maintain software
- **Platform as a Service (PaaS)**
 - Users are given software and hardware automatically
- **Software as a Service (SaaS)**
 - All software and hardware is transparent
 - User only knows their own access point



Service Models of Cloud Computing



1. Infrastructure as a service (IaaS)

Infrastructure as a Service (IaaS) helps in delivering computer infrastructure on an external basis for supporting operations. Generally, IaaS provides services to networking equipment, devices, databases, and web servers.

Infrastructure as a Service (IaaS) helps large organizations, and large enterprises in managing and building their IT platforms. This infrastructure is flexible according to the needs of the client.

Advantages of IaaS

- IaaS is cost-effective as it eliminates capital expenses.
- IaaS cloud provider provides better security than any other software.
- IaaS provides remote access.

Disadvantages of IaaS

- In IaaS, users have to secure their own data and applications.
- Cloud computing is not accessible in some regions of the World.



Platform as a service (PaaS)

Platform as a Service (PaaS) is a type of cloud computing that helps developers to build applications and services over the Internet by providing them with a platform.

PaaS helps in maintaining control over their business applications.

Advantages of PaaS

- PaaS is simple and very much convenient for the user as it can be accessed via a web browser.
- PaaS has the capabilities to efficiently manage the lifecycle.

Disadvantages of PaaS

- PaaS has limited control over infrastructure as they have less control over the environment and are not able to make some customizations.
- PaaS has a high dependence on the provider.



Software as a service (SaaS)

Software as a Service (SaaS) is a type of cloud computing model that is the work of delivering services and applications over the Internet. The SaaS applications are called Web-Based Software or Hosted Software.

SaaS has around 60 percent of cloud solutions and due to this, it is mostly preferred by companies.

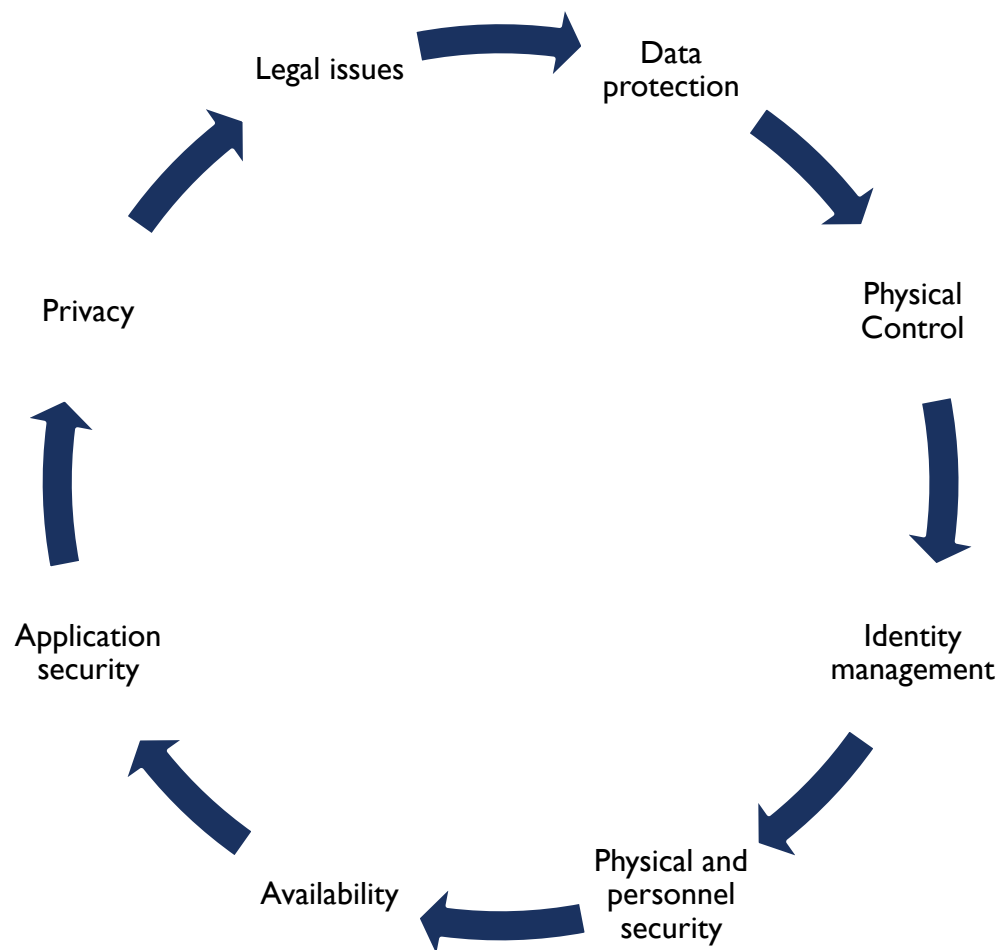
Advantages of SaaS

- SaaS can access app data from anywhere on the Internet.
- SaaS provides easy access to features and services.

Disadvantages of SaaS

- SaaS solutions have limited customization, which means they have some restrictions within the platform.
- SaaS has little control over the data of the user.
- SaaS are generally cloud-based, they require a stable internet connection for proper working.

SECURITY AND PRIVACY



CLOUD CLIENTS



GOOGLE CHROME
BOOK



WEB BROWSERS



COMPANY SPECIFIC
APPLICATIONS

ADVANTAGES



CLOUD EVERYDAY
USE



TO SAVE HUGE
AMOUNTS OF DATA



EASIER TO MAINTAIN
INFORMATION



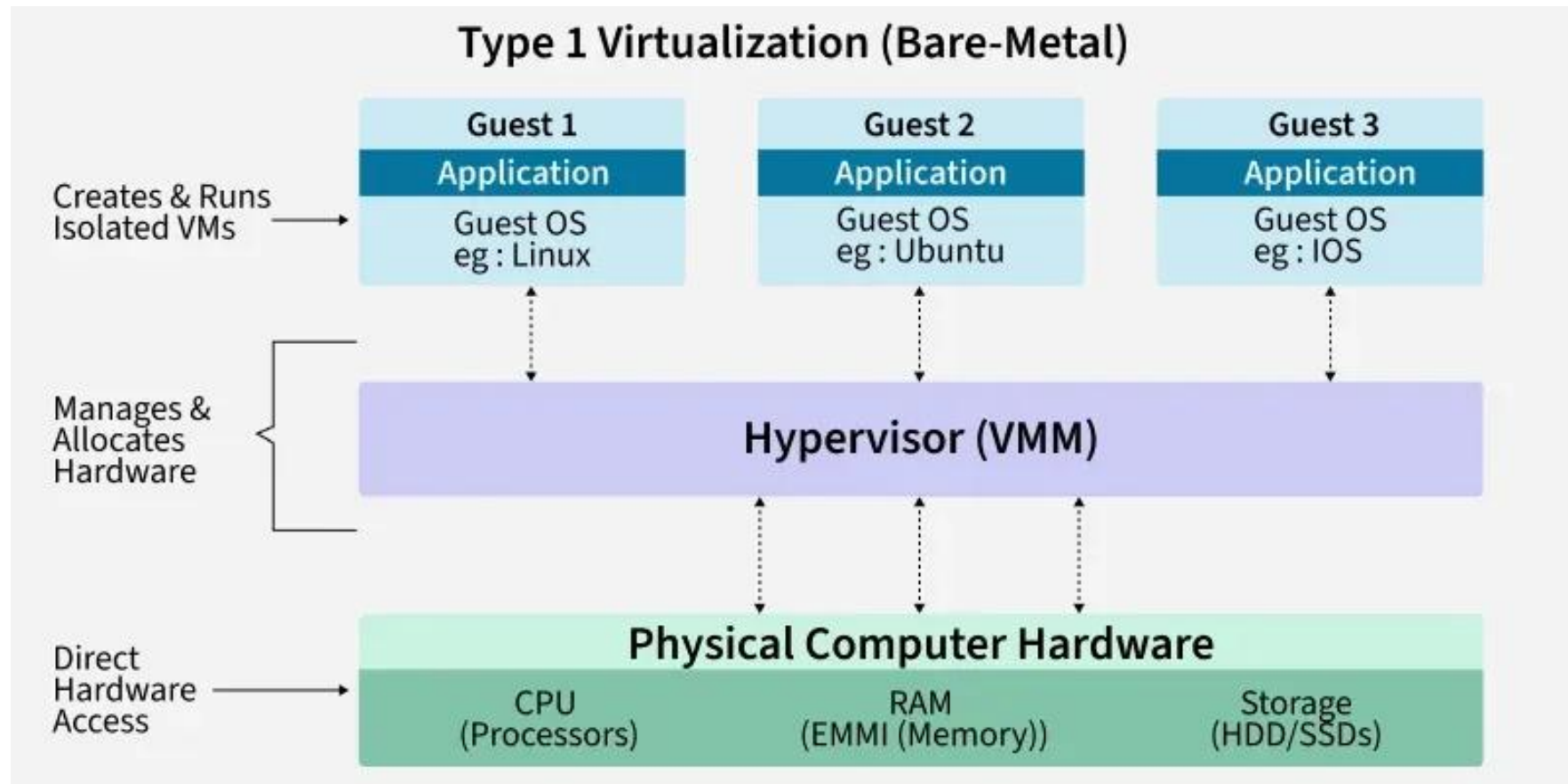
MAKES SECURITY EASY



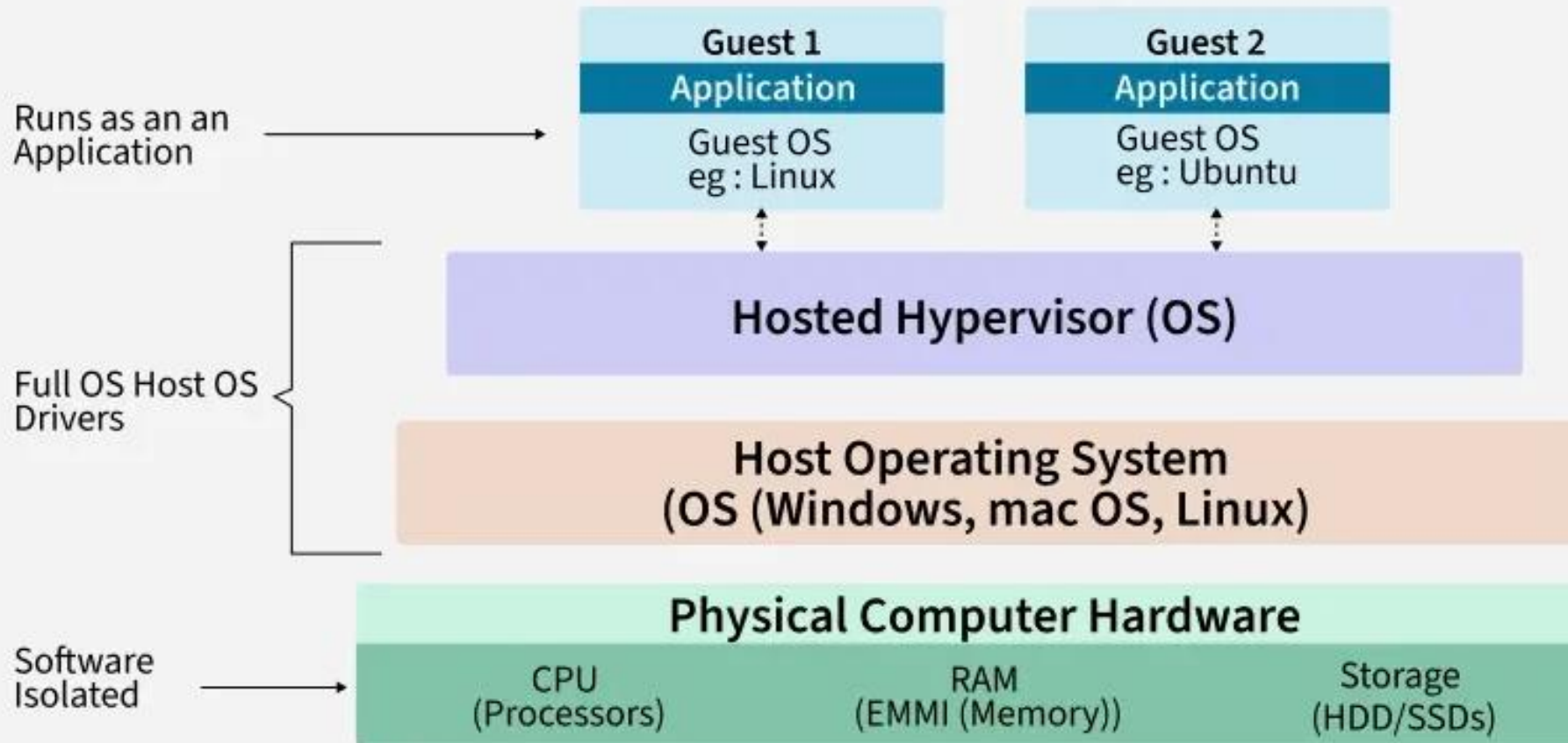
MAINTAINABILITY
AND SUSTAINABILITY
ARE BETTER

Feature	Virtualization	Cloud Computing
Definition	Technology that creates virtual versions of hardware.	Service that delivers shared computing resources via the internet.
Relationship	The Tool: Virtualization is the software technology.	The Service: Cloud Computing is the result or service built using virtualization.
Ownership	You usually own/manage the hardware (unless using Cloud).	You rent resources; the provider owns the hardware.
Focus	Maximizing hardware efficiency.	Maximizing user agility and scalability.

VIRTUALIZATION



Type 2 Virtualization (Hosted)



THE TWO TYPES OF HYPERVISORS

Type 1: Bare-Metal Hypervisor

- Installed directly on the physical hardware. There is no host Operating System.
- High Performance. Direct access to hardware resources.
- Use Case: Enterprise Data Centers, Cloud Providers (AWS EC2, VMWare ESXi, Microsoft Hyper-V).

Type 2: Hosted Hypervisor

- Installed as an application *on top* of an existing OS (like Windows or macOS).
- Lower Performance . Requests must pass through the Host OS first.
- Use Case: Personal use, testing labs (Oracle VirtualBox, VMWare Workstation).

Types of Virtualization

```
graph TD; A[Types of Virtualization] --> B[Application Virtualization]; A --> C[Network Virtualization]; A --> D[Desktop Virtualization]; A --> E[Storage Virtualization]; A --> F[Server Virtualization]; A --> G[Data Virtualization];
```

Application
Virtualization

Network
Virtualization

Desktop
Virtualization

Storage
Virtualization

Server
Virtualization

Data
Virtualization



Data Virtualization: It aggregates the heterogeneous data from a different source to a single logical or virtual volume of data.

Processor virtualization: It is the process of abstracting physical processor to the pool of virtualization processor.

Memory virtualization: It is the process of providing virtual main memory to the VMS that are abstracted from the physical main memory.

Storage : It is a form of resource virtualization where a multiple physical storage is abstracted as a multiple logical storage.

Network: It is the process of abstracting physical networking components to form a virtual network.

Application: It allows the users to access the virtual instance of the centrally hosted application without any installation.

DATA VIRTUALIZATION – OVERVIEW

- Data Virtualization is a technique that provides a **unified, logical view of data**
- Allows access to data **without physically moving or copying it**
- Abstracts data location, format, and access mechanisms
- Enables **real-time data integration** across multiple sources
- Common in cloud, big data, and enterprise systems

Need for Data Virtualization

- Data is distributed across **databases, cloud storage, APIs, and legacy systems**
- Traditional ETL causes **latency, duplication, and storage overhead**
- Real-time analytics require **fresh and consistent data**
- Reduces **data integration complexity**
- Supports agile and scalable data architectures



Architecture of Data Virtualization

•Data Sources

- Relational databases, NoSQL, data lakes, cloud storage

•Data Virtualization Layer

- Metadata management
- Query optimization and federation

•Access Layer

- SQL, REST APIs, BI tools, applications

•Security & Governance

- Access control, masking, auditing

Key Features and Technologies

- Logical Data Abstraction
- Query Federation & Optimization
- Real-Time Data Access
- Caching and Performance Acceleration
- Data Transformation & Enrichment
- Tools/Platforms:
 - Denodo, Red Hat Data Virtualization
 - IBM Cloud Pak for Data
 - Azure Data Virtualization (via Synapse)



Benefits and Use Cases

•Benefits

- Faster time-to-insight
- Reduced data duplication and storage cost
- Improved data consistency
- Simplified data management

•Use Cases

- Business intelligence and reporting
- Cloud–on-premise data integration
- Data lakes and hybrid cloud analytics
- API-based data access for applications



Approaches to Virtualization:

1. **Full Virtualization:** It is the process of completely abstracting the underlying physical infrastructure with binary translation and direct execution.
2. **Para Virtualization or OS assisted Virtualization:** It partially abstracts the underlying infrastructure with hypercalls.
3. **Hardware assisted Virtualization:** It eliminates the overhead of binary translation and hypercalls, where the hardware vendors themselves support Virtualization.

- 
1. Using Microsoft App-V or Citrix to run Microsoft Excel on an iPad. The app runs on a server, but the user sees it on their tablet.
 2. AWS VPC (Virtual Private Cloud), you can create subnets and route tables in software, without touching a physical router.
 3. Amazon WorkSpaces, an employee can log in from a Chromebook, but sees a full high-power Windows 11 desktop running in the cloud
 4. SAN (Storage Area Network) or Amazon S3, you can see a single "bucket" or drive, but the data is physically spread across hundreds of hard drives.
 5. Running a Web Server (Linux), a Database (Windows), and a Mail Server (Linux) all on one physical machine using VMware vSphere.
 6. Denodo or Oracle Data Service. A dashboard queries "Sales Data," and the virtualization layer pulls it from both an old SQL database and a new Cloud Data Lake instantly.



UNIT-II

CLOUD SERVICE PLATFORMS

CLOUD COMPUTING SERVICE PLATFORMS

- Cloud Service Platforms provide **on-demand computing resources** over the internet
- Enable **scalable, flexible, and cost-effective** application development and deployment
- Abstract underlying hardware and infrastructure complexities
- Support **pay-as-you-go** and **resource pooling** models
- Examples: **AWS, Microsoft Azure, Google Cloud Platform**

Key Components of Cloud Service Platforms

- **Compute Services** – Virtual Machines, Containers, Serverless
- **Storage Services** – Object, Block, and File Storage
- **Networking Services** – Virtual Networks, Load Balancers, Gateways
- **Security & Identity** – IAM, Encryption, Key Management
- **Management Tools** – Monitoring, Logging, Auto-scaling



Service Models Supported by Platforms

•Infrastructure as a Service (IaaS)

- Virtual servers, storage, networking
- Example: Amazon EC2, Azure Virtual Machines

•Platform as a Service (PaaS)

- Application runtime and development tools
- Example: Google App Engine, Azure App Services

•Software as a Service (SaaS)

- Ready-to-use applications
- Example: Salesforce, Google Workspace

Popular Cloud Service Platforms

•Amazon Web Services (AWS)

- Largest market share, wide service portfolio

•Microsoft Azure

- Strong enterprise and hybrid cloud integration

•Google Cloud Platform (GCP)

- Advanced data analytics and AI services

•IBM Cloud & Oracle Cloud

- Enterprise-focused and industry-specific solutions



Benefits and Use Cases

•Benefits

- High availability and fault tolerance
- Rapid deployment and scalability
- Reduced infrastructure and maintenance costs

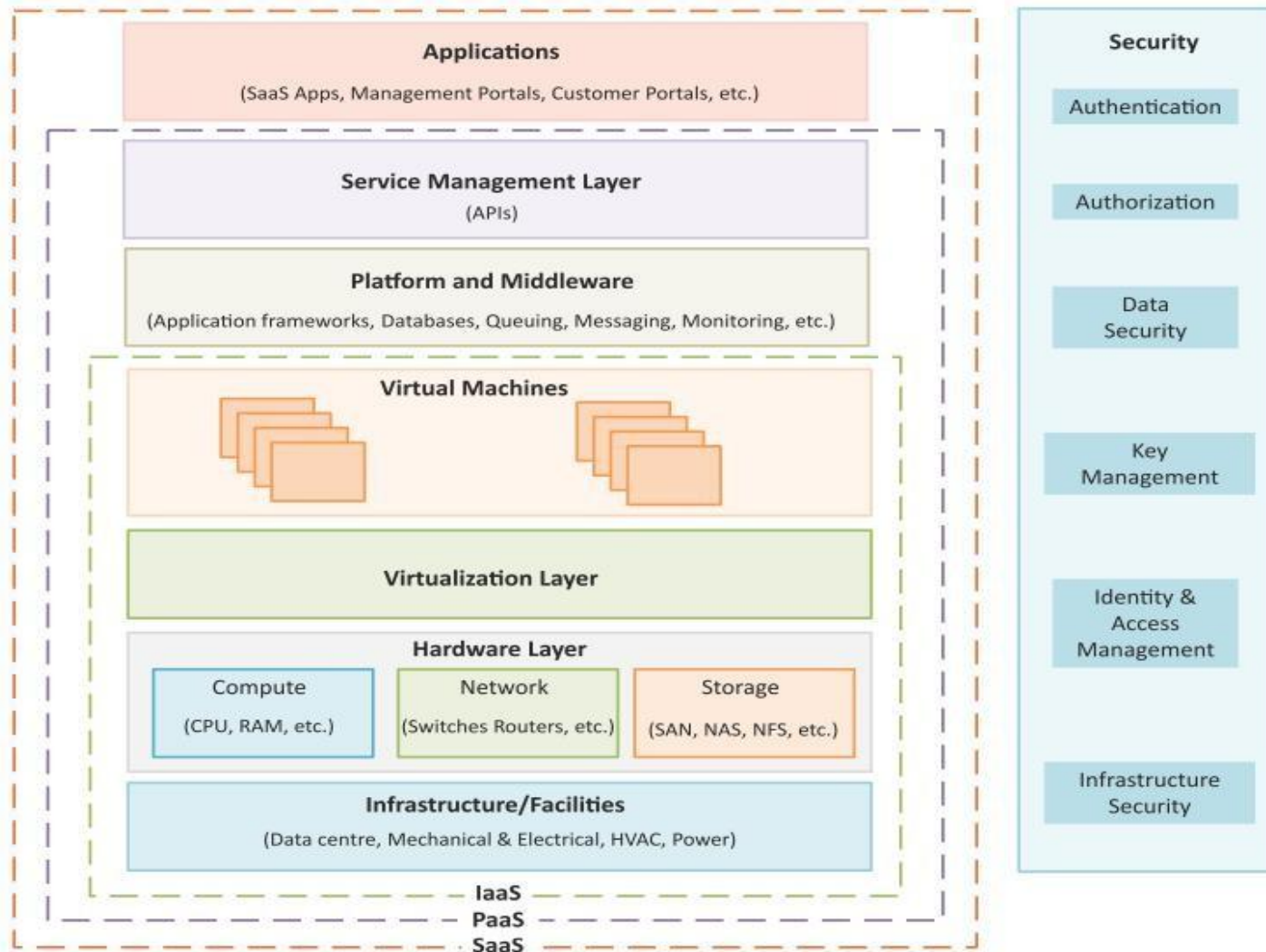
•Use Cases

- Web and mobile application hosting
- Big data analytics and AI/ML
- Disaster recovery and backup
- DevOps and CI/CD pipelines

LIST OF SERVICES

- Compute Services
- Storage Services
- Database Services
- Application Services
- Content Delivery Services
- Analytics Services
- Deployment & Management Services
- Identity & Access Management Services

CLOUD REFERENCE MODEL



CLOUD REFERENCE MODEL

- Infrastructure & Facilities Layer
 - Includes the physical infrastructure such as datacenter facilities, electrical and mechanical equipment, etc.
- Hardware Layer
 - Includes physical compute, network and storage hardware.
- Virtualization Layer
 - Partitions the physical hardware resources into multiple virtual resources that enabling pooling of resources.
- Platform & Middleware Layer
 - Builds upon the IaaS layers below and provides standardized stacks of services such as database service, queuing service, application frameworks and run-time environments, messaging services, monitoring services, analytics services, etc.
- Service Management Layer
 - Provides APIs for requesting, managing and monitoring cloud resources.
- Applications Layer
 - Includes SaaS applications such as Email, cloud storage application, productivity applications, management portals, customer self-service portals, etc.

COMPUTE SERVICES

- Compute services provide dynamically scalable compute capacity in the cloud.
- Compute resources can be provisioned on-demand in the form of virtual machines. Virtual machines can be created from standard images provided by the cloud service provider or custom images created by the users.
- Compute services can be accessed from the web consoles of these services that provide graphical user interfaces for provisioning, managing and monitoring these services.
- Cloud service providers also provide APIs for various programming languages that allow developers to access and manage these services programmatically.

COMPUTE SERVICES – AMAZON EC2

The screenshot displays the Amazon EC2 console interface. The top navigation bar includes the AWS logo, 'Services' and 'Edit' dropdowns, the region 'Oregon', and a 'Help' link. The left sidebar contains a navigation menu with categories: EC2 Dashboard (Events, Tags), INSTANCES (Instances, Spot Requests, Reserved Instances), IMAGES (AMIs, Bundle Tasks), ELASTIC BLOCK STORE (Volumes, Snapshots), and NETWORK & SECURITY (Security Groups, Elastic IPs, Placement Groups, Load Balancers, Key Pairs, Network Interfaces).

The main content area is titled 'Resources' and shows a summary of EC2 resources in the US West (Oregon) region:

- 0 Running Instances
- 0 Elastic IPs
- 0 Volumes
- 0 Snapshots
- 0 Key Pairs
- 0 Load Balancers
- 0 Placement Groups
- 12 Security Groups

Below the resource summary is a blue box with the text: 'Optimize your resources' cost, performance and security with AWS Trusted Advisor'. A 'Hide' link is present on the right side of this box.

The 'Create Instance' section provides instructions: 'To start using Amazon EC2 you will want to launch a virtual server, known as an Amazon EC2 instance.' and includes a 'Launch Instance' button.

A note states: 'Note: Your instances will launch in the US West (Oregon) region'.

The 'Service Health' section shows the status of the service in the US West (Oregon) region:

- Service Status:** US West (Oregon): This service is operating normally (indicated by a green checkmark).
- Availability Zone Status:**
 - us-west-2a: Availability zone is operating normally (indicated by a green checkmark).
 - us-west-2b: Availability zone is operating normally (indicated by a green checkmark).

The 'Scheduled Events' section shows 'No events' for the US West (Oregon) region.

The right sidebar contains 'Account Attributes' (Supported Platforms: EC2-VPC, Default VPC: vpc-8d4117) and 'Additional Information' (Getting Started Guide, Documentation, All EC2 Resources, Forums, Pricing, Contact Us). Below this is the 'Popular AMIs on AWS Marketplace' section, featuring 'Debian GNU/Linux' (Provided by Debian, Rating: 5 stars, Free Software, pay only for AWS usage, View all Operating Systems) and 'Couchbase Server - Community Edition' (Provided by Couchbase, Rating: 5 stars, Free Software, pay only for AWS usage, View all Databases).

COMPUTE SERVICES – AMAZON EC2

- Amazon Elastic Compute Cloud (EC2) is a compute service provided by Amazon.
- Launching EC2 Instances
 - To launch a new instance click on the launch instance button. This will open a wizard where you can select the Amazon machine image (AMI) with which you want to launch the instance. You can also create their own AMIs with custom applications, libraries and data. Instances can be launched with a variety of operating systems.
- Instance Sizes
 - When you launch an instance you specify the instance type (micro, small, medium, large, extra-large, etc.), the number of instances to launch based on the selected AMI and availability zones for the instances.
- Key-pairs
 - When launching a new instance, the user selects a key-pair from existing keypairs or creates a new keypair for the instance. Keypairs are used to securely connect to an instance after it launches.
- Security Groups
 - The security groups to be associated with the instance can be selected from the instance launch wizard. Security groups are used to open or block a specific network port for the launched instances.

COMPUTE SERVICES – GOOGLE COMPUTE ENGINE

Google Cloud Console

Cloud Project ID: cloud Compute Engine

NEW INSTANCE

Create a new Instance

Name

Description

Tags

Metadata

Location and Resources

Zone

Machine Type

Boot Source

Image

Additional Disks

Networking

Network

External IP

Summary

myinstance

My instance

debian-7-wheezy-v20130723

us-central1-b

1 vCPU, 3.75 GB RAM

default

Note: per-minute charges will begin now

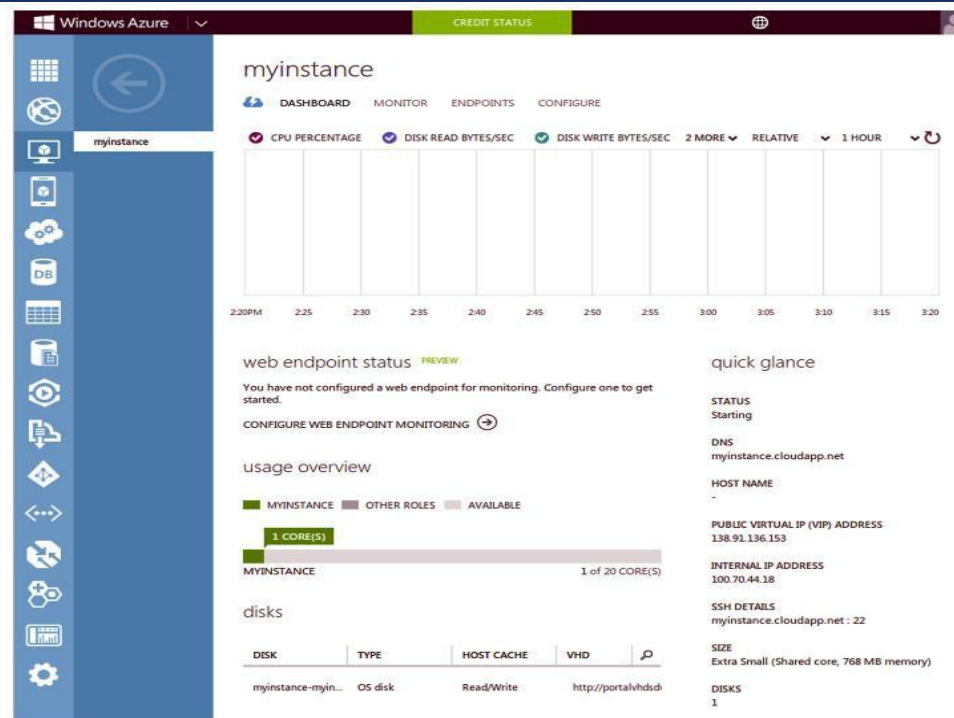
Create Discard

Equivalent REST or command line

- Network Options

Network option allows you to control the traffic to and from the instances. By default, traffic between instances in the same network, over any port and any protocol and incoming SSH connections from anywhere are enabled.
- Google Compute Engine is a compute service provided by Google.
- Launching Instances
 - To create a new instance, the user selects an instance machine type, a zone in which the instance will be launched, a machine image for the instance and provides an instance name, instance tags and meta-data.
- Disk Resources
 - Every instance is launched with a disk resource. Depending on the instance type, the disk resource can be a scratch disk space or persistent disk space. The scratch disk space is deleted when the instance terminates. Whereas, persistent disks live beyond the life of an instance.

COMPUTE SERVICES – WINDOWS AZURE VMS

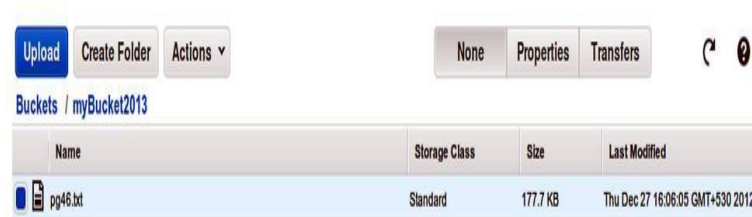


- Windows Azure Virtual Machines is the compute service from Microsoft.
- Launching Instances:
 - To create a new instance, you select the instance type and the machine image.
 - You can either provide a user name and password or upload a certificate file for securely connecting to the instance.
 - Any changes made to the VM are persistently stored and new VMs can be created from the previously stored machine images.

STORAGE SERVICES

- Cloud storage services allow storage and retrieval of any amount of data, at any time from anywhere on the web.
- Most cloud storage services organize data into buckets or containers.
- Scalability
 - Cloud storage services provide high capacity and scalability. Objects upto several tera-bytes in size can be uploaded and multiple buckets/containers can be created on cloud storages.
- Replication
 - When an object is uploaded it is replicated at multiple facilities and/or on multiple devices within each facility.
- Access Policies
 - Cloud storage services provide several security features such as Access Control Lists (ACLs), bucket/container level policies, etc. ACLs can be used to selectively grant access permissions on individual objects. Bucket/container level policies can also be defined to allow or deny permissions across some or all of the objects within a single bucket/container.
- Encryption
 - Cloud storage services provide Server Side Encryption (SSE) options to encrypt all data stored in the cloud storage.
- Consistency
 - Strong data consistency is provided for all upload and delete operations. Therefore, any object that is uploaded can be immediately downloaded after the upload is complete.

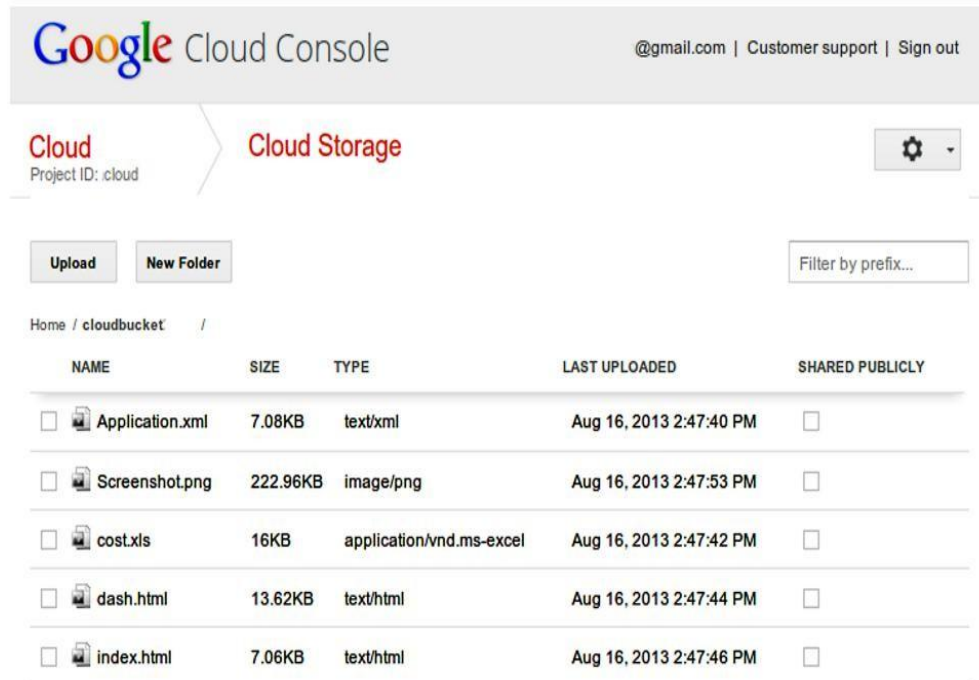
STORAGE SERVICES – AMAZON S3



- Amazon Simple Storage Service(S3) is an online cloud-based data storage infrastructure for storing and retrieving any amount of data.
- S3 provides highly reliable, scalable, fast, fully redundant and affordable storage infrastructure.
- Buckets
 - Data stored on S3 is organized in the form of buckets. You must create a bucket before you can store data on S3.
- Uploading Files to Buckets
 - S3 console provides simple wizards for creating a new bucket and uploading files.
 - You can upload any kind of file to S3.
 - While uploading a file, you can specify the redundancy and encryption options and access permissions.

STORAGE SERVICES – GOOGLE CLOUD STORAGE

- GCS is the Cloud storage service from Google
- Buckets
 - Objects in GCS are organized into buckets.
- Access Control Lists
 - ACLs are used to control access to objects and buckets. ACLs can be configured to share objects and buckets with the entire world, a Google group, a Google-hosted domain, or specific Google account holders.



Google Cloud Console

@gmail.com | Customer support | Sign out

Cloud Project ID: cloud > Cloud Storage

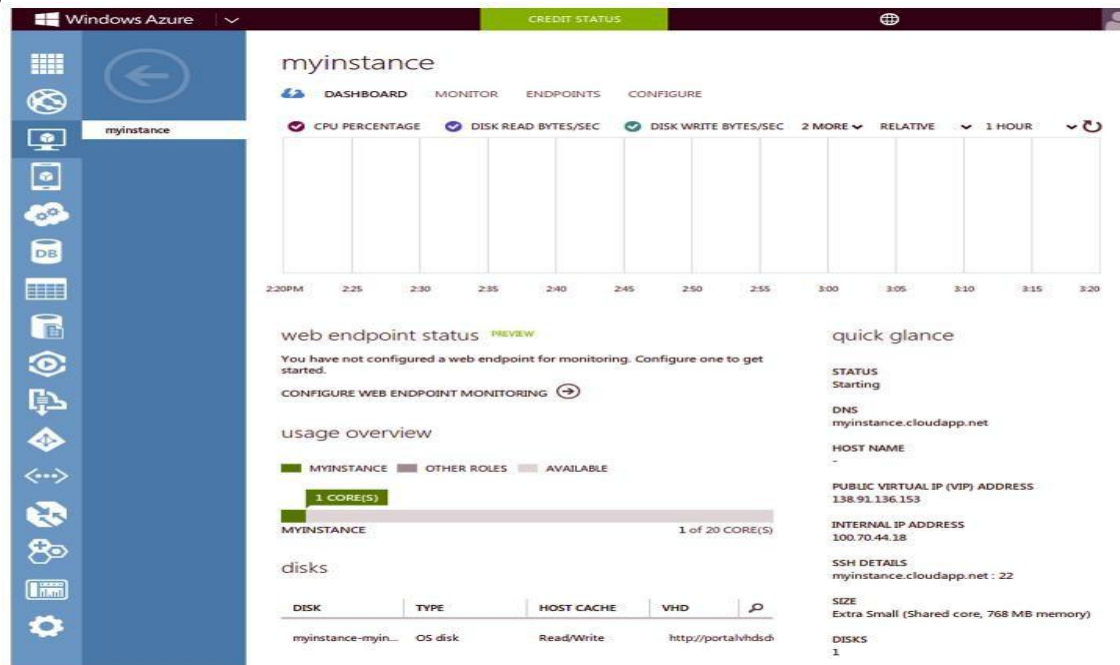
Upload New Folder Filter by prefix...

Home / cloudbucket /

	NAME	SIZE	TYPE	LAST UPLOADED	SHARED PUBLICLY
☐	 Application.xml	7.08KB	text/xml	Aug 16, 2013 2:47:40 PM	☐
☐	 Screenshot.png	222.96KB	image/png	Aug 16, 2013 2:47:53 PM	☐
☐	 cost.xls	16KB	application/vnd.ms-excel	Aug 16, 2013 2:47:42 PM	☐
☐	 dash.html	13.62KB	text/html	Aug 16, 2013 2:47:44 PM	☐
☐	 index.html	7.06KB	text/html	Aug 16, 2013 2:47:46 PM	☐

STORAGE SERVICES – WINDOWS AZURE STORAGE

- Windows Azure Storage is the cloud storage service from Microsoft.
- Windows Azure Storage provides various storage services such as blob storage service, table service and queue service.
- Blob storage service
 - The blob storage service allows storing unstructured binary data or binary large objects (blobs).
 - Blobs are organized into containers.
 - Block blobs - can be subdivided into some number of blocks. If a failure occurs while transferring a block blob, retransmission can resume with the most recent block rather than sending the entire blob again.
 - Page blobs - are divided into number of pages and are designed for random access. Applications can read and write individual pages at random in a page blob.



DATABASE SERVICES

- Cloud database services allow you to set-up and operate relational or non-relational databases in the cloud.
- Relational Databases
 - Popular relational databases provided by various cloud service providers include MySQL, Oracle, SQL Server, etc.
- Non-relational Databases
 - The non-relational (No-SQL) databases provided by cloud service providers are mostly proprietary solutions.
- Scalability
 - Cloud database services allow provisioning as much compute and storage resources as required to meet the application workload levels. Provisioned capacity can be scaled-up or down. For read-heavy workloads, read-replicas can be created.
- Reliability
 - Cloud database services are reliable and provide automated backup and snapshot options.
- Performance
 - Cloud database services provide guaranteed performance with options such as guaranteed input/output operations per second (IOPS) which can be provisioned upfront.
- Security
 - Cloud database services provide several security features to restrict the access to the database instances and stored data, such as network firewalls and authentication mechanisms.

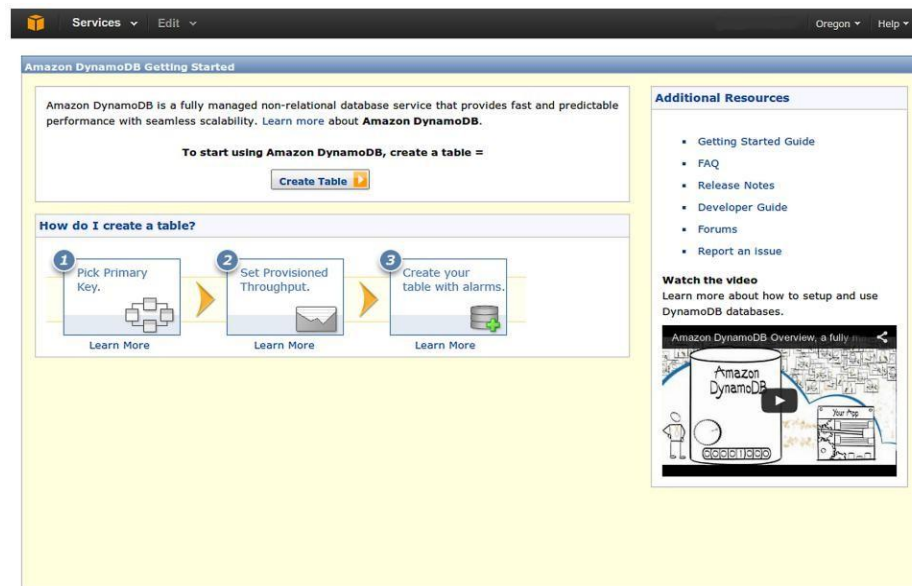
DATABASE SERVICES – AMAZON RDS

- Amazon Relational Database Service (RDS) is a web service that makes it easy to setup, operate and scale a relational database in the cloud.
- Launching DB Instances
 - The console provides an instance launch wizard that allows you to select the type of database to create (MySQL, Oracle or SQL Server) database instance size, allocated storage, DB instance identifier, DB username and password. The status of the launched DB instances can be viewed from the console.
- Connecting to a DB Instance
 - Once the instance is available, you can note the instance end point from the instance properties tab. This end point can then be used for securely connecting to the instance.

The screenshot shows the Amazon RDS console dashboard. At the top, there's a navigation bar with 'Services' and 'Edit' dropdowns, and the region 'Oregon'. On the left, a sidebar lists navigation options: RDS Dashboard, Database, Instances, Reserved Purchases, Snapshots, Parameter Groups, Option Groups, Subnet Groups, Events, and Event Subscriptions. The main content area has a welcome message: 'Welcome to the new RDS console interface. Learn more about the updates and send us your feedback.' Below this, there are three sections: 'Resources' (listing DB Instances (0), DB Snapshots (0), DB Parameter Groups (0), Reserved DB Purchases (0), Recent Events (0), Supported Platforms VPC, and Default Network vpc-8d4117e5), 'Create Instance' (with a 'Launch a DB Instance' button and a note about the US West (Oregon) region), and 'Service Health' (showing 'Amazon Relational Database Service (Oregon)' as 'Service is operating normally'). On the right, there's an 'Additional Information' section with links to 'Getting Started with RDS', 'Overview and Features', 'Documentation', 'Articles and Tutorials', 'Data import guide for MySQL', 'Data import guide for Oracle', 'Data import guide for SQL Server', 'Pricing', and 'Forums'. Below that is a 'Related Services' section for 'Amazon ElastiCache'.

DATABASE SERVICES – AMAZON DYNAMODB

- Amazon DynamoDB is the non-relational (No-SQL) database service from Amazon.
- Data Model
 - The DynamoDB data model includes include tables, items and attributes.
 - A table is a collection of items and each item is a collection of attributes.
 - To store data in DynamoDB you have to create a one or more tables and specify how much throughput capacity you want to provision and reserve for reads and writes.
- Fully Managed Service
 - DynamoDB is a fully managed service that automatically spreads the data and traffic for the stored tables over a number of servers to meet the throughput requirements specified by the users.
- Replication : All stored data is automatically replicated across multiple availability zones to provide data durability.



DATABASE SERVICES – GOOGLE CLOUD SQL

- Google SQL is the relational database service from Google.
- Google Cloud SQL service allows you to host MySQL databases in the Google's cloud.
- Launching DB Instances
 - You can create new database instances from the console and manage existing instances. To create a new instance you select a region, database tier, billing plan and replication mode.
- Backups : You can schedule daily backups for your Google Cloud SQL instances, and also restore backed-up databases.
- Replication
 - Cloud SQL provides both synchronous or asynchronous geographic replication and the ability to import/ export databases.

The screenshot shows the Google Cloud Console interface for creating a new Cloud SQL instance. The breadcrumb navigation at the top indicates the path: Cloud > Cloud SQL > cloud:mydbinstance. On the left, there is a sidebar with a 'NEW INSTANCE' button and links for 'Summary' and 'Operations Log'. The main content area is titled 'Create a New Cloud SQL Instance' and contains several sections: 'Instance ID' with a text field containing 'mydbinstance'; 'Region' with a dropdown menu set to 'United States'; 'Resources and Billing' section which includes a 'Tier' dropdown set to 'D0 — 128M RAM', a 'Billing Plan' section with radio buttons for 'Per Use' (selected) and 'Package', and an 'Options' section with a 'Backup Window' dropdown set to '7:30 AM — 11:30 AM' and a 'Replication' section with radio buttons for 'Synchronous' (selected) and 'Asynchronous'. A 'Summary' panel on the right provides a overview of the configuration: instance ID 'mydbinstance', region 'United States', tier 'D0' (50.025 per hour - 128M RAM), billing plan 'Per Use', backup window 'Backups at 7:30 AM — 11:30 AM', and replication mode 'Synchronous replication'. At the bottom of the summary panel are 'Confirm' and 'Cancel' buttons.

DATABASE SERVICES – GOOGLE CLOUD DATASTORE

- Google Cloud Datastore is a fully managed non-relational database from Google.
- Cloud Datastore offers ACID transactions and high availability of reads and writes.
- Data Model
 - The Cloud Datastore data model consists of entities. Each entity has one or more properties (key-value pairs) which can be of one of several supported data types, such as strings and integers. Each entity has a kind and a key. The entity kind is used for categorizing the entity for the purpose of queries and the entity key uniquely identifies the entity.

The screenshot shows the Google Cloud Console interface for creating a new entity in Cloud Datastore. The top navigation bar includes the Google Cloud Console logo and user links. The left sidebar shows navigation options: Stats, Query, and Indexes. The main content area is titled 'Cloud Datastore' and features a 'CREATE ENTITY' button. Below this, the 'Create new entity' form is displayed. It includes a 'Namespace' dropdown set to 'Default', a 'Kind' dropdown set to 'New kind', and a text input field containing 'Person'. Below these, there is a section for defining properties. A property named 'Name' is shown with a data type of 'a string', an 'Indexed' checkbox, and a value of 'Nav Kaur'. At the bottom, there are 'Create entity' and 'Cancel' buttons.

Google Cloud Console @gmail.com | Customer support | Sign out

Cloud Project ID: cloud Cloud Datastore ⚙️

Stats CREATE ENTITY

Query

Indexes

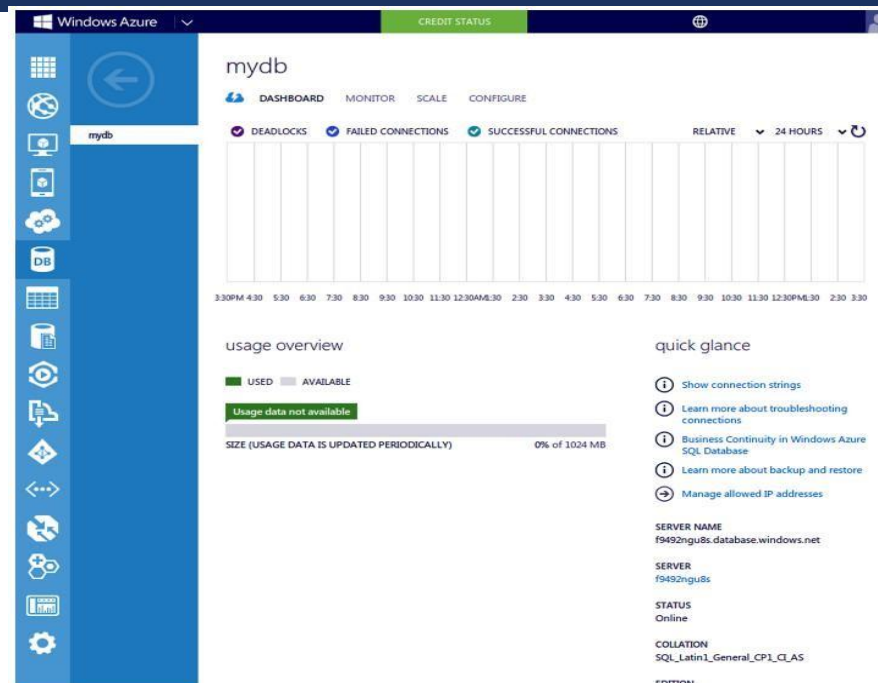
Create new entity

Namespace Default Kind New kind Person

Name a string Indexed Nav Kaur + -

Create entity Cancel

DATABASE SERVICES –WINDOWS AZURE SQL DB



- Windows Azure SQL Database is the relational database service from Microsoft.
- Azure SQL Database is based on the SQL server, but it does not give each customer a separate instance of SQL server.
- Multi-tenant Service
 - SQL Database is a multi-tenant service, with a logical SQL Database server for each customer.

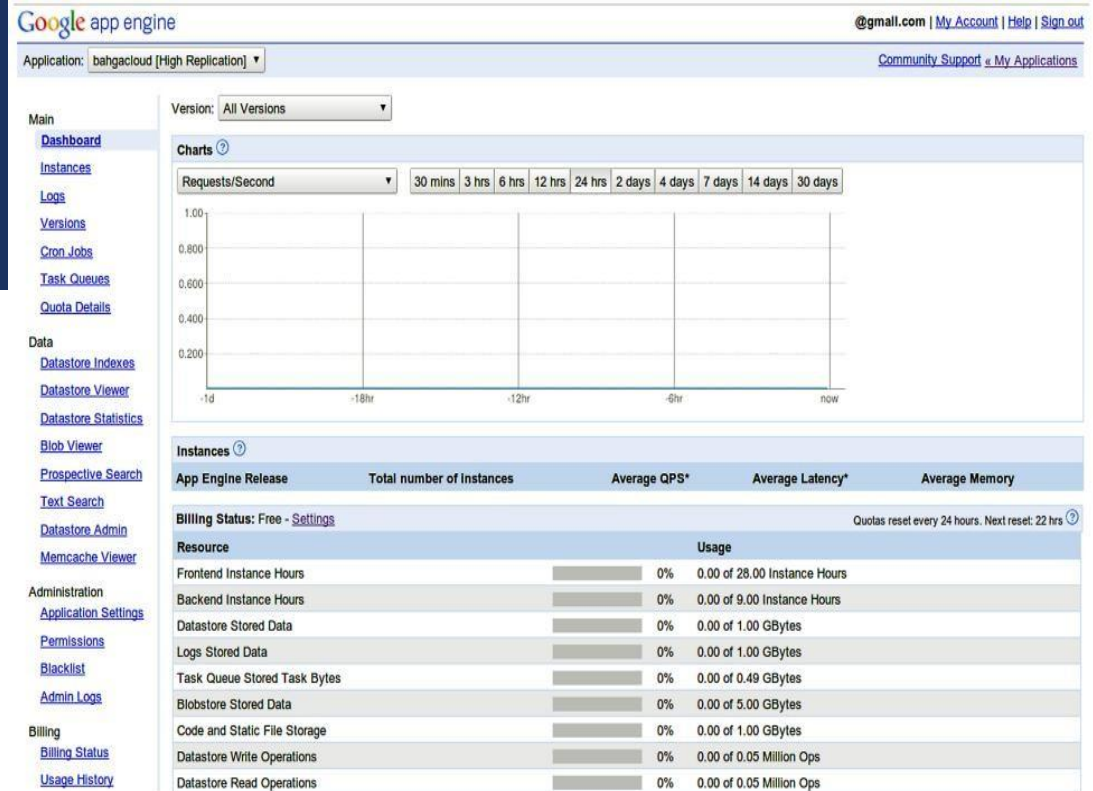
DATABASE SERVICES – WINDOWS AZURE TABLE SERVICE

- Windows Azure Table Service is a non-relational (No-SQL) database service from Microsoft.
- Data Model
 - The Azure Table Service data model consists of tables having multiple entities.
 - Tables are divided into some number of partitions, each of which can be stored on a separate machine.
 - Each partition in a table holds a specified number of entities, each containing as many as 255 properties.
 - Each property can be one of the several supported data types such as integers and strings.
- No Fixed Schema
 - Tables do not have a fixed schema and different entities in a table can have different properties.

APPLICATION RUNTIMES & FRAMEWORKS

- Cloud-based application runtimes and frameworks allow developers to develop and host applications in the cloud.
- Support for various programming languages
 - Application runtimes provide support for programming languages (e.g., Java, Python, or Ruby).
- Resource Allocation
 - Application runtimes automatically allocate resources for applications and handle the application scaling, without the need to run and maintain servers.

GOOGLE APP ENGINE



- Google App Engine is the platform-as-a-service (PaaS) from Google, which includes both an application runtime and web frameworks.
- Runtimes
 - App Engine provides runtime environments for Java, Python, PHP and Go programming language.
- Sandbox
 - Applications run in a secure sandbox environment isolated from other applications.
 - The sandbox environment provides a limited access to the underlying operating system.

GOOGLE APP ENGINE

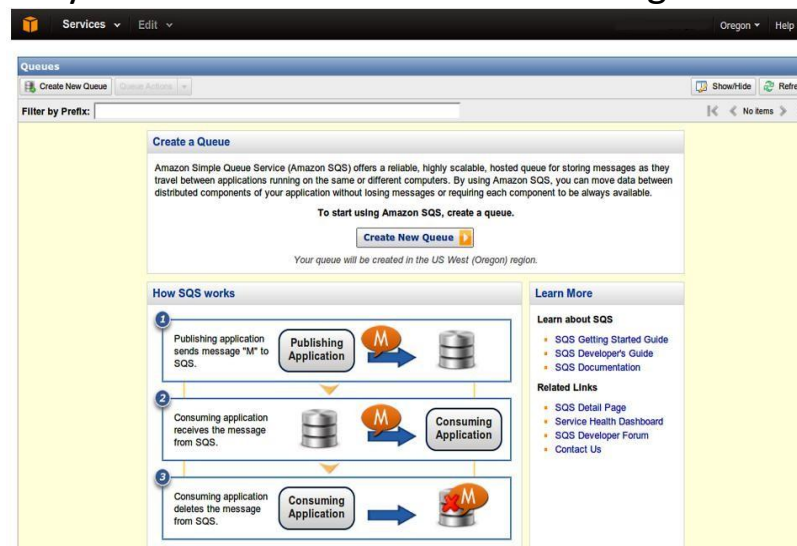
- Web Frameworks
 - App Engine provides a simple Python web application framework called webapp2. App Engine also supports any framework written in pure Python that speaks WSGI, including Django, CherryPy, Pylons, web.py, and web2py.
- Datastore
 - App Engine provides a no-SQL data storage service.
- Authentication
 - App Engine applications can be integrated with Google Accounts for user authentication.
- URL Fetch service
 - URL Fetch service allows applications to access resources on the Internet, such as web services or other data.
- Other services
 - Email service
 - Image Manipulation service
 - Memcache
 - Task Queues
 - Scheduled Tasks service

WINDOWS AZURE WEB SITES

- Windows Azure Web Sites is a Platform-as-a-Service (PaaS) from Microsoft.
- Azure Web Sites allows you to host web applications in the Azure cloud.
- Shared & Standard Options.
 - In the shared option, Azure Web Sites run on a set of virtual machines that may contain multiple web sites created by multiple users.
 - In the standard option, Azure Web Sites run on virtual machines (VMs) that belong to an individual user.
- Azure Web Sites supports applications created in ASP.NET, PHP, Node.js and Python programming languages.
- Multiple copies of an application can be run in different VMs, with websites automatically load balancing requests across them.

QUEUING SERVICES - AMAZON SIMPLE QUEUE SERVICE

- Amazon Simple Queue Service (SQS) is a queuing service from Amazon.
- Short Messages
 - SQS is a distributed queue that supports messages of up to 256 KB in size.
- Multiple Writers/Readers
 - SQS supports multiple writers and readers and locks messages while they are being processed.
- High Availability
 - To ensure high availability for delivering messages, SQS service trade-offs on the first in, first out capability and does not guarantee that messages will be delivered in FIFO order.
 - Applications that require FIFO ordering of messages can place additional sequencing information in each message so that they can be re-ordered after retrieving from a queue.



QUEUING SERVICES - GOOGLE TASK QUEUE SERVICE

- Google Task Queues service is a queuing service from Google and is a part of the Google App Engine platform.
- Task queues allow applications to execute tasks in background.
- Tasks
 - Task is a unit of work to be performed by an application. The task objects consist of application-specific URL with a request handler for the task, and an optional data payload that parameterizes the task.
- Push Queue
 - Push Queue is the default queue that processes tasks based on the processing rate configured in the queue definition.
- Pull Queue
 - Pull Queues allow task consumers to lease a specific number of tasks for a specific duration. The tasks are processed and deleted before the lease ends.

QUEUING SERVICES - WINDOWS AZURE QUEUE SERVICE

- Windows Azure Queue service is a queuing service from Microsoft.
- Azure Queue service allows storing large numbers of messages that can be accessed from anywhere in the world via authenticated calls using HTTP or HTTPS.
- Short Messages
 - The size of a single message can be up to 64KB.

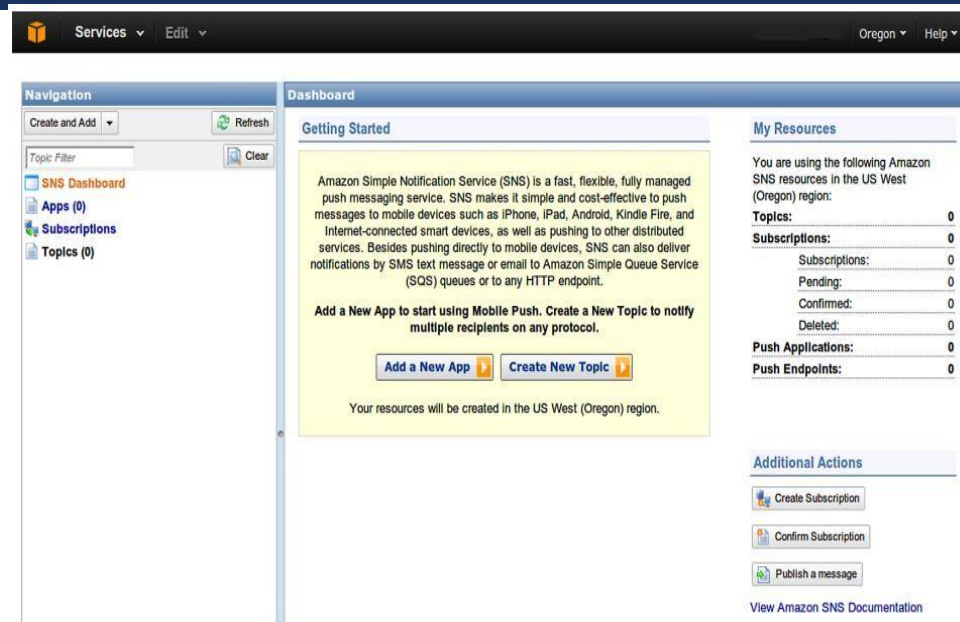
EMAIL SERVICES

- Cloud-based email services allow applications hosted in the cloud to send emails.
- Amazon Simple Email Service (SES)
 - Amazon Simple Email Service is bulk and transactional email-sending service from Amazon
 - SES is an outbound-only email-sending service that allows applications hosted in the Amazon cloud to send emails such as marketing emails, transactional emails and other types of correspondence
 - To ensure high email deliverability, SES uses content filtering technologies to scan the outgoing email messages
 - SES service can be accessed and used from the SES console, the Simple Mail Transfer Protocol (SMTP) interface, or the SES API
- Google Email Service
 - Google Email service is part of the Google App Engine platform that allows App Engine applications to send email messages on behalf of the app's administrators, and on behalf of users with Google Accounts.
 - App Engine apps can also receive emails. Apps send messages using the Mail service and receive messages in the form of HTTP requests initiated by App Engine and posted to the app.

NOTIFICATION SERVICES

- Cloud-based notification services or push messaging services allow applications to push messages to internet connected smart devices such as smartphones, tablets, etc.
- Push messaging services are based on publish-subscribe model in which consumers subscribe to various topics/channels provided by a publisher/producer.
- Whenever new content is available on one of those topics/channels, the notification service pushes that information out to the consumer.
- Push notifications are used for such smart devices as they help in displaying the latest information while remaining energy efficient.
- Consumer applications on such devices can increase their consumer engagement with the help of push notifications.

NOTIFICATION SERVICES - AMAZON SIMPLE NOTIFICATION SERVICE



- Amazon Simple Notification Service(SNS) is a push messaging service from Amazon.
- SNS has two types of clients:
 - Publishers
 - Publishers communicate asynchronously with subscribers by producing and sending messages to topics. A topic is a logical access point and a communication channel.
 - Subscribers.
 - Subscribers are the consumers who subscribe to topics to receive notifications.
- SNS can deliver notifications as SMS, email or to SQS queues, or any HTTP endpoint.

GOOGLE CLOUD MESSAGING

- Google Cloud Messaging for Android provides push messaging for Android devices.
- GCM allows applications to send data from the application servers to their users' Android devices, and also to receive messages from devices on the same connection.
- Notifying Android Apps
 - GCM is useful for notifying applications on Android devices that there is new data to be fetched from the application servers.
- Short Messages
 - GCM supports messages with payload data upto 4 KB.
- Send-to-Sync
 - GCM provides a 'send-to-sync' message capability that can be used to inform an application to sync data from the server.
- GCM for Chrome
 - Google Cloud Messaging for Chrome is another notification service from Google that allows messages to be delivered from the cloud to apps and extensions running in Chrome.

WINDOWS AZURE NOTIFICATION HUBS

- Windows Azure Notification Hubs is a push notification service from Microsoft.
- Common Interface
 - Provides a common interface to send notifications to all major mobile platforms including Windows Store/Windows Phone 8, iOS, and Android.
- Platform Notification Systems
 - Platform specific infrastructures called Platform Notification Systems (PNS) are used to deliver notification messages.
 - Devices register their PNS handles with the Notification Hub.
 - Each notification hub contains credentials for each supported PNS.
 - These credentials are used to connect to the PNSs and send push notifications to the applications.

MEDIA SERVICES

- Cloud service providers provide various types of media services that can be used by applications for manipulating, transforming or transcoding media such as images, videos, etc.
- Amazon Elastic Transcoder
 - Amazon Elastic Transcoder is a cloud-based video transcoding service from Amazon.
 - Elastic Transcoder can be used to convert video files from their source format into various other formats that can be played on devices such as desktops, mobiles, tablets, etc.
- Google Images Manipulation Service
 - Google Images Manipulation service is a part of the Google App Engine platform. Image Manipulation service provides the capability to resize, crop, rotate, flip and enhance images.
- Windows Azure Media Services
 - Windows Azure Media Services provides the various media services such as encoding & format conversion, content protection and on-demand and live streaming capabilities.

CONTENT DELIVERY SERVICES

- Cloud-based content delivery service include Content Delivery Networks (CDNs).
- CDN is a distributed system of servers located across multiple geographic locations to serve content to end- users with high availability and high performance.
- CDNs are useful for serving static content such as text, images, scripts, etc., and streaming media.
- CDNs have a number of edge locations deployed in multiple locations, often over multiple backbones.
- Requests for static for streaming media content that is served by a CDN are directed to the nearest edge location.
- Amazon CloudFront
 - Amazon CloudFront is a content delivery service from Amazon. CloudFront can be used to deliver dynamic, static and streaming content using a global network of edge locations.
- Windows Azure Content Delivery Network
 - Windows Azure Content Delivery Network (CDN) is the content delivery service from Microsoft.

ANALYTICS SERVICES

- Cloud-based analytics services allow analyzing massive data sets stored in the cloud either in cloud storages or in cloud databases using programming models such as MapReduce.
- Amazon Elastic MapReduce
 - Amazon Elastic MapReduce is the MapReduce service from Amazon based the Hadoop framework running on Amazon EC2 and S3
 - EMR supports various job types such as Custom JAR, Hive program, Streaming job, Pig programs and Hbase
- Google MapReduce Service
 - Google MapReduce Service is a part of the App Engine platform and can be accessed using the Google MapReduce API.
- Google BigQuery
 - Google BigQuery is a service for querying massive datasets. BigQuery allows querying datasets using SQL-like queries.
- Windows Azure HDInsight
 - Windows Azure HDInsight is an analytics service from Microsoft. HDInsight deploys and provisions Hadoop clusters in the Azure cloud and makes Hadoop available as a service.

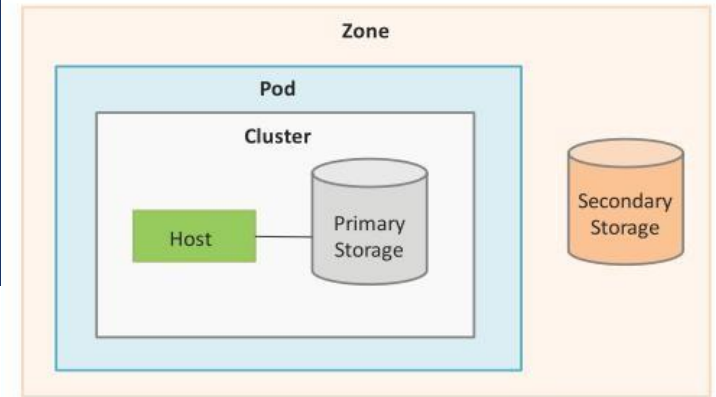
DEPLOYMENT & MANAGEMENT SERVICES

- Cloud-based deployment & management services allow you to easily deploy and manage applications in the cloud. These services automatically handle deployment tasks such as capacity provisioning, load balancing, auto-scaling, and application health monitoring.
- Amazon Elastic Beanstalk
 - Amazon provides a deployment service called Elastic Beanstalk that allows you to quickly deploy and manage applications in the AWS cloud.
 - Elastic Beanstalk supports Java, PHP, .NET, Node.js, Python, and Ruby applications.
 - With Elastic Beanstalk you just need to upload the application and specify configuration settings in a simple wizard and the service automatically handles instance provisioning, server configuration, load balancing and monitoring.
- Amazon CloudFormation
 - Amazon CloudFormation is a deployment management service from Amazon.
 - With CloudFront you can create deployments from a collection of AWS resources such as Amazon Elastic Compute Cloud, Amazon Elastic Block Store, Amazon Simple Notification Service, Elastic Load Balancing and Auto Scaling.
 - A collection of AWS resources that you want to manage together are organized into a stack.

IDENTITY & ACCESS MANAGEMENT SERVICES

- Identity & Access Management (IDAM) services allow managing the authentication and authorization of users to provide secure access to cloud resources.
- Using IDAM services you can manage user identifiers, user permissions, security credentials and access keys.
- Amazon Identity & Access Management
 - AWS Identity and Access Management (IAM) allows you to manage users and user permissions for an AWS account.
 - With IAM you can manage users, security credentials such as access keys, and permissions that control which AWS resources users can access.
 - Using IAM you can control what data users can access and what resources users can create.
 - IAM also allows you to control creation, rotation, and revocation security credentials of users.
- Windows Azure Active Directory
 - Windows Azure Active Directory is an Identity & Access Management Service from Microsoft.
 - Azure Active Directory provides a cloud-based identity provider that easily integrates with your on-premises active directory deployments and also provides support for third party identity providers.
 - With Azure Active Directory you can control access to your applications in Windows Azure.

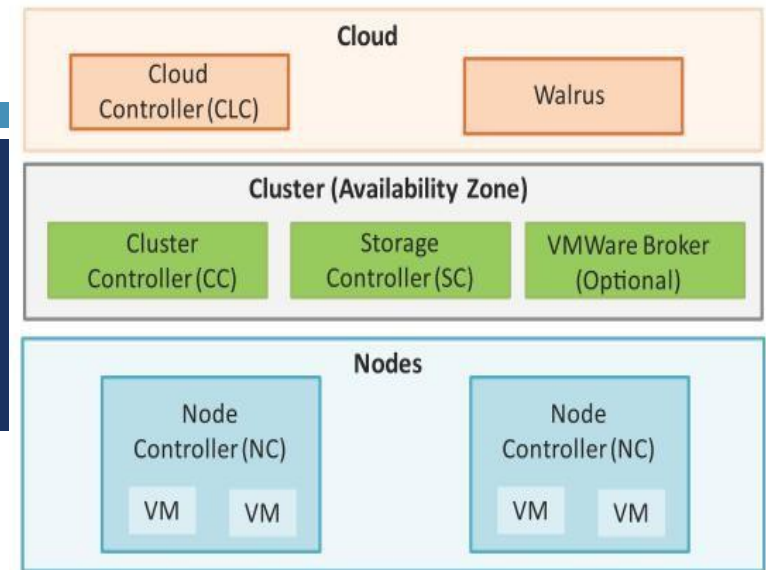
OPEN SOURCE PRIVATE CLOUD SOFTWARE - CLOUDSTACK



- Apache CloudStack is an open source cloud software that can be used for creating private cloud offerings.
- CloudStack manages the network, storage, and compute nodes that make up a cloud infrastructure.
- A CloudStack installation consists of a Management Server and the cloud infrastructure that it manages.
- Zones
 - The Management Server manages one or more zones where each zone is typically a single datacenter.
- Pods
 - Each zone has one or more pods. A pod is a rack of hardware comprising of a switch and one or more clusters.
- Cluster
 - A cluster consists of one or more hosts and a primary storage. A host is a compute node that runs guest virtual machines.
- Primary Storage
 - The primary storage of a cluster stores the disk volumes for all the virtual machines running on the hosts in that cluster.
- Secondary Storage
 - Each zone has a secondary storage that stores templates, ISO images, and disk volume snapshots.

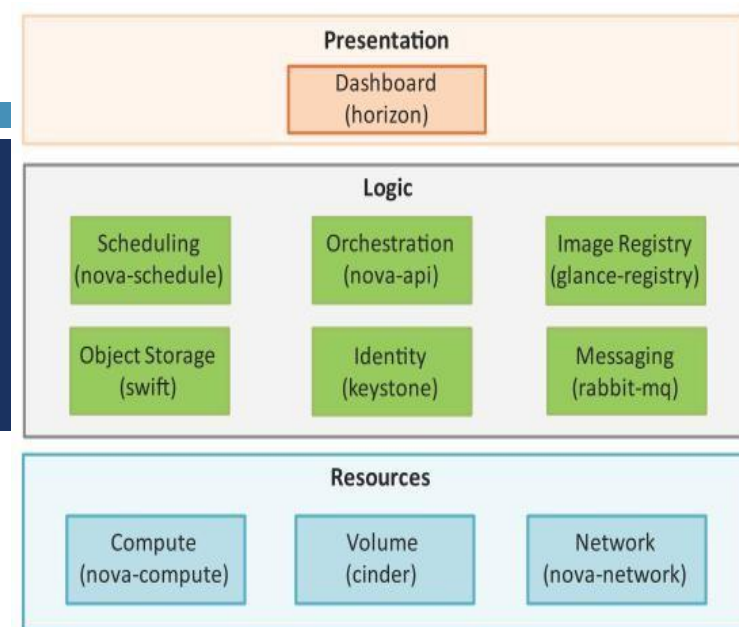
OPEN SOURCE PRIVATE CLOUD SOFTWARE - EUCLYPTUS

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OPEN SOURCE PRIVATE CLOUD SOFTWARE - OPENSTACK

- Eucalyptus is an open source private cloud software for building private and hybrid clouds that are compatible with Amazon Web Services (AWS) APIs.
- Node Controller hosts the VM instances & manages the virtual network endpoints.
- The cluster-level (availability-zone) consists of three components
 - Cluster Controller - which manages the virtual machines and is the front-end for a cluster.
 - Storage Controller – which manages the Eucalyptus block volumes and snapshots to the instances within its specific cluster. SC is equivalent to AWS Elastic Block Store (EBS).
 - VMWare Broker - which is an optional component that provides an AWS-compatible interface for VMware environments.
- At the cloud-level there are two components:
 - Cloud Controller - which provides an administrative interface for cloud management and performs high-level resource scheduling, system accounting, authentication and quota management.
 - Walrus - which is equivalent to Amazon S3 and serves as a persistent storage to all of the virtual machines in the Eucalyptus cloud. Walrus can be used as a simple Storage-as-a-Service



FURTHER READING

- Amazon Elastic Compute Cloud, <http://aws.amazon.com/ec2>
- Google Compute Engine, <https://developers.google.com/compute/>
- Windows Azure, <http://www.windowsazure.com/>
- Google App Engine, <http://appengine.google.com>
- Google App Engine, <https://developers.google.com/appengine/>
- Google Cloud Storage, <https://developers.google.com/storage/>
- Google BigQuery, <https://developers.google.com/bigquery/>
- Google Cloud Datastore, <http://developers.google.com/datastore/>
- Google Cloud SQL, <https://developers.google.com/cloud-sql/>
- CloudStack, <http://cloudstack.apache.org>
- Eucalyptus, <http://www.eucalyptus.com>
- OpenStack, <http://www.openstack.org>